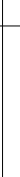
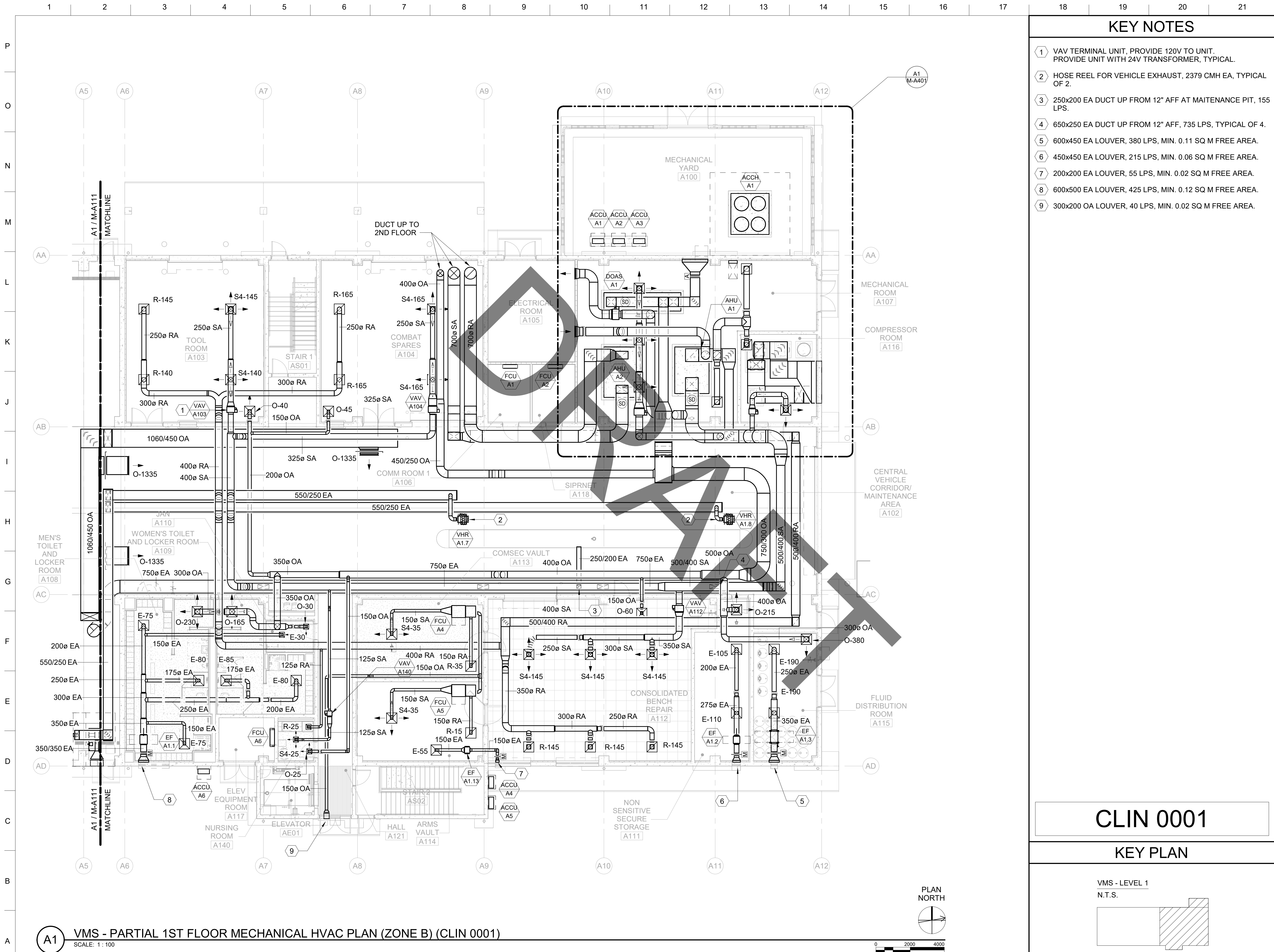




INTERMEDIATE DESIGN



 US Army Corps of Engineers®		DATE	
ISSUE DATE:		MARK	
DESIGNED BY: T. TSANG J. W. YU R. BALIGNASAY OKINAWA AREA OFFICE APO AP 96376-5132		SOLICITATION NO.: CONTRACT NO.: W912HV-17-D-0003	
CHECKED BY: O. CRASTREE		SUBMITTED BY:	
SIZE: A		SIZE: A	
US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96376-5132		DESIGN PARTNERS INCORPORATED 1580 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814	



KEY NOTES

- VAV TERMINAL UNIT, PROVIDE 120V TO UNIT. PROVIDE UNIT WITH 24V TRANSFORMER, TYPICAL.
- HOSE REEL FOR VEHICLE EXHAUST, 2379 CMH EA, TYPICAL OF 2.
- 250x200 EA DUCT UP FROM 12" AFF AT MAITENANCE PIT, 155 LPS.
- 650x250 EA DUCT UP FROM 12" AFF, 735 LPS, TYPICAL OF 4.
- 600x450 EA LOUVER, 380 LPS, MIN. 0.11 SQ M FREE AREA.
- 450x450 EA LOUVER, 215 LPS, MIN. 0.06 SQ M FREE AREA.
- 200x200 EA LOUVER, 55 LPS, MIN. 0.02 SQ M FREE AREA.
- 600x500 EA LOUVER, 425 LPS, MIN. 0.12 SQ M FREE AREA.
- 300x200 OA LOUVER, 40 LPS, MIN. 0.02 SQ M FREE AREA.

US Army Corps of Engineers	
DATE	
DESCRIPTION	
MARK	

DESIGNED BY: I. TSANG	ISSUE DATE:
DRAWN BY: R. BALINASAY	SOLICITATION NO.:
CHECKED BY: W. J. J. J. J.	CONTRACT NO.:
SUBMITTED BY:	
SIZE: ANSI D	

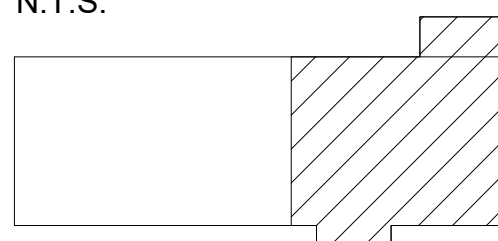
US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96376-5132	DESIGN PARTNERS INCORPORATED 1580 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814
FY20 MCA PN7833 VEHICLE MAINTENANCE SHOP KADENA AIR BASE, OKINAWA, JAPAN	
VMS - PARTIAL 1ST FLOOR MECHANICAL HVAC PLAN (ZONE B) (CLIN 0001)	

SHEET ID
M-A112

CLIN 0001

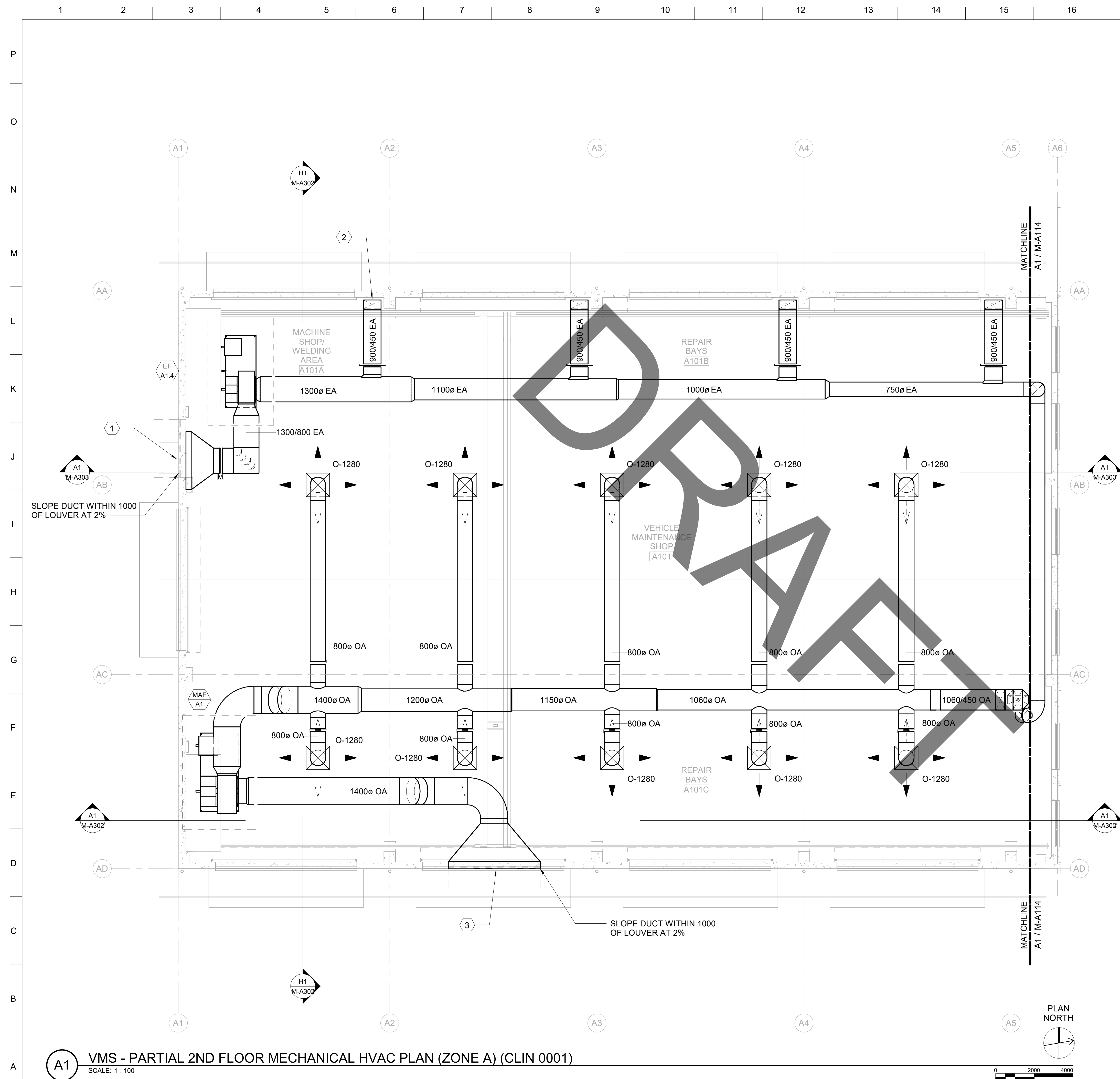
KEY PLAN

VMS - LEVEL 1
N.T.S.



INTERMEDIATE DESIGN

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KEY NOTES

- | | |
|---|---|
| 1 | 3000x3000 EA LOUVER, 12,965 LPS, MINIMUM 3.65 SQ M FREE AREA. |
| 2 | EA DUCT UP FROM 12" AFF, 2470 LPS, TYPICAL OF 4. |
| 3 | 4750x2000 OA LOUVER AT 8600MM SILL HEIGHT, 16,815 LPS, MINIMUM 4.73 SQ M FREE AREA. |



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[illegible]

US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96376-5132	DESIGNED BY: I TSANG	ISSUE DATE:
DESIGN PARTNERS INCORPORATED 1580 KALANANAKU AVENUE, SUITE 1100 HONOLULU, HAWAII 96814	DRAWN BY: RICHARD L. RAY	SOLICITATION NO.:
	CHECKED BY: O. CRABTREE	CONTRACT NO.:
	SUBMITTED BY:	W912N-V-17-D-0003
	SIZE:	

FY20 MCA PN76833 VEHICLE MAINTENANCE SHOP
KADENA AIR BASE, OKINAWA, JAPAN

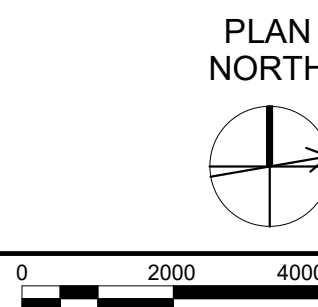
SHEET ID

M-A113

INTERMEDIATE DESIGN

Plot Date: 12/15/2025 3:35:11 PM

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- [illegible]

FY20 MCA PN77833 VEHICLE MAINTENANCE SHOP
KADENA AIR BASE, OKINAWA, JAPAN

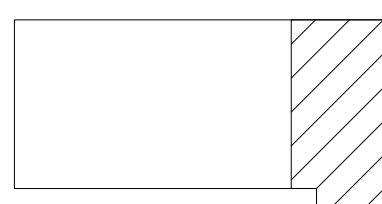
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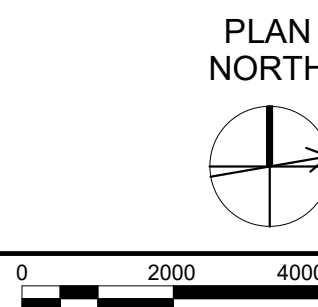
KEY PLAN

VMS - LEVEL 2
N.T.S.



SHEET ID

M-A114



- 1 RG AND RE PIPING UP TO SECOND FLOOR FCU-A3.
- 2 20MM CD TO TERMINATE AT MECH ROOM FLOOR SINK.
- 3 32MM CD TO TERMINATE AT JAN MOP SINK.
- 4 20MM CD TO CONNECT TO SINK TAILPIPE.

[illegible]

US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96376-5132	DESIGNED BY:	ISSUE DATE:
	DRAWN BY: KINAWA AREA OFFICE	SOLICITATION NO.:
	CHECKED BY: O. CRABTREE	CONTRACT NO. W812HW-17-D-0003
DESIGN PARTNERS, INCORPORATED 1560 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814	SUBMITTED BY:	
	SIZE: ANSI D	

FY20 MCA PN77833 VEHICLE MAINTENANCE SHOP
KADENA AIR BASE, OKINAWA, JAPAN

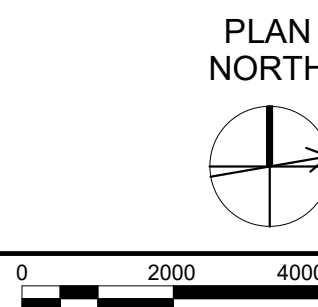
SHEET ID

M-A115

INTERMEDIATE DESIGN

Plot Date: 12/15/2025 3:35:17 PM File Path:

\\users\hoffmange\Documents\511758 TIME MECH 11/2022 Hoffmange.vnd



- 1 RG AND RL PIPING DN TO FIRST FLOOR CEILING
- 2 32MM CD TO TERMINATE AT JAN MOP SINK.

[illegible]

US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96376-5132	DESIGNED BY: I. TSANG	ISSUE DATE:
	DRAWN BY: T. KAWASAKI CHECKED BY: O. CRABTREE SUBMITTED BY:	SOLICITATION NO.: CONTRACT NO.: W812HW-17-D-0003
DESIGN PARTNERS, INCORPORATED 1560 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814	SIZE: ANSI D	

FY20 MCA PN7833 VEHICLE MAINTENANCE SHOP
KADENA AIR BASE, OKINAWA, JAPAN

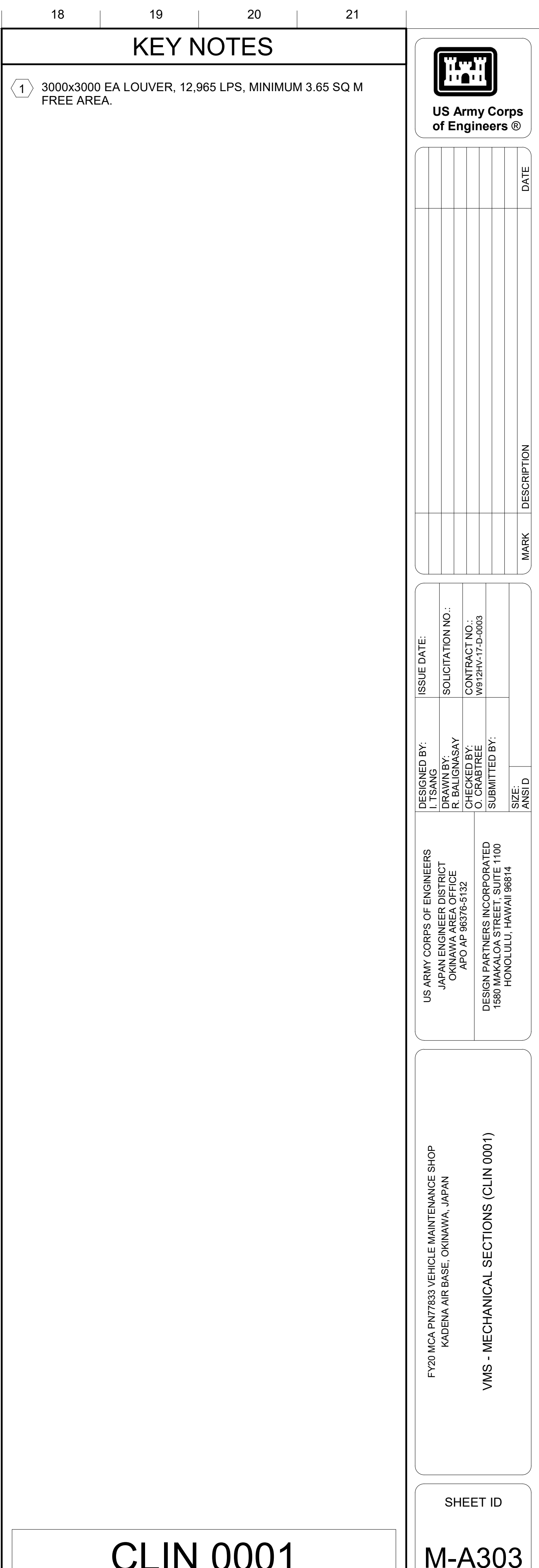
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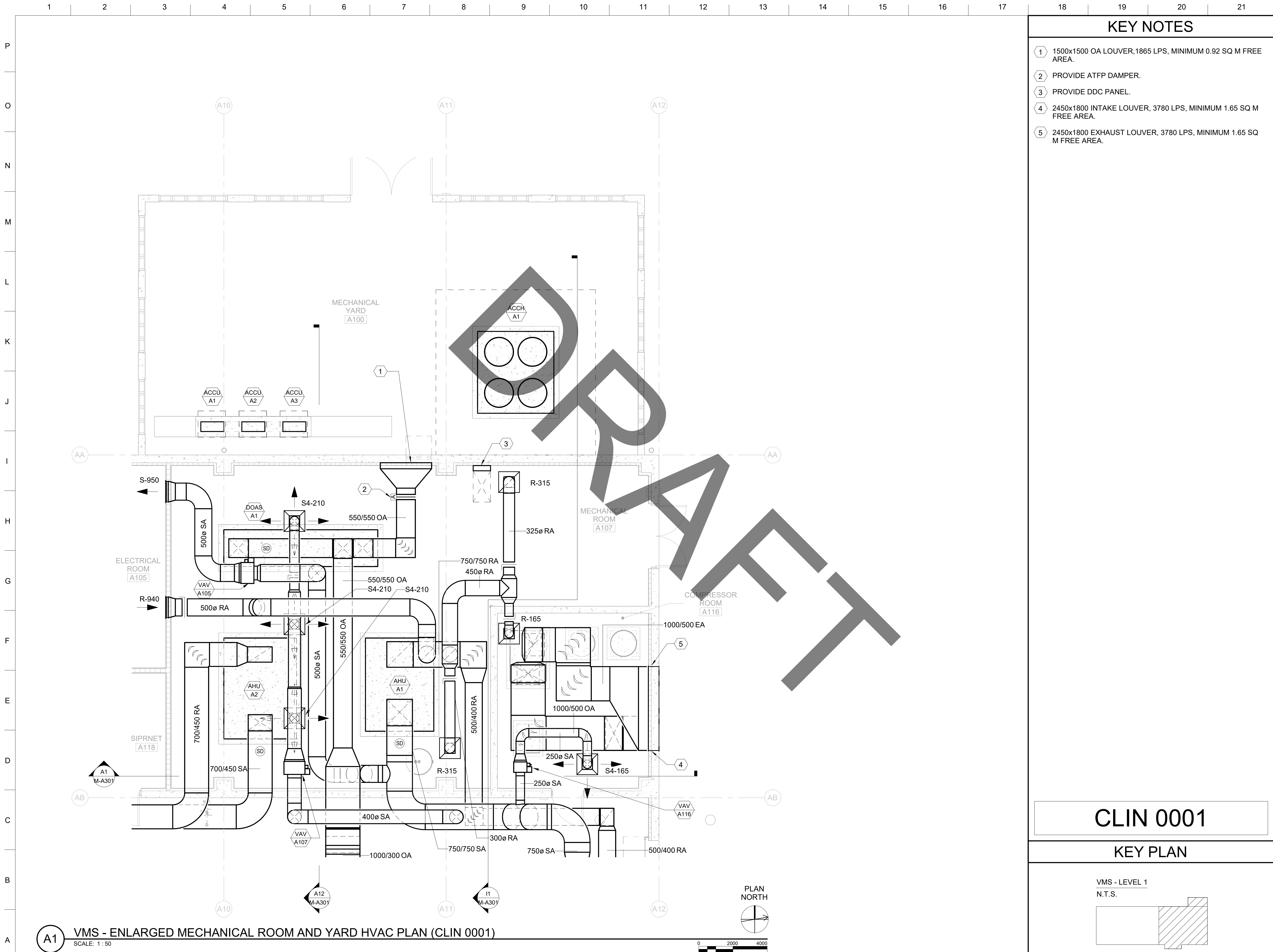
M-A116

INTERMEDIATE DESIGN

Plot Date: 12/15/2025 3:35:18 PM File Path:

SAJID JALIL, *Ph.D., is an associate professor at the Department of Mechanical Engineering, University of Malakand, Pakistan. He has published several research papers in international journals and conferences. His research interests include finite element analysis, optimization, and mechanical design.*





A1 VMS - ENLARGED MECHANICAL ROOM AND YARD HVAC PLAN (CLIN 0001)
SCALE: 1 : 50

KEY NOTES

- 1500x1500 OA LOUVER, 1865 LPS, MINIMUM 0.92 SQ M FREE AREA.
- PROVIDE ATFP DAMPER.
- PROVIDE DDC PANEL.
- 2450x1800 INTAKE LOUVER, 3780 LPS, MINIMUM 1.65 SQ M FREE AREA.
- 2450x1800 EXHAUST LOUVER, 3780 LPS, MINIMUM 1.65 SQ M FREE AREA.



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MARK	DESCRIPTION	DATE

DESIGNED BY: I. TSANG	ISSUE DATE:
DRAWN BY: R. BALIGASAY	SOLICITATION NO.:
CHECKED BY: WJZ/RY-17-D-0003	CONTRACT NO.:
SUBMITTED BY:	
SIZE: ANSI D	

US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA AREA OFFICE
APO AF 96376-5132
DESIGN PARTNERS INCORPORATED
1580 MAKALOA STREET, SUITE 1100
HONOLULU, HAWAII 96814

FY20 MCA PN7833 VEHICLE MAINTENANCE SHOP KADENA AIR BASE, OKINAWA, JAPAN	VMS - ENLARGED MECHANICAL ROOM AND YARD HVAC PLAN (CLIN 0001)
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SHEET ID

M-A401

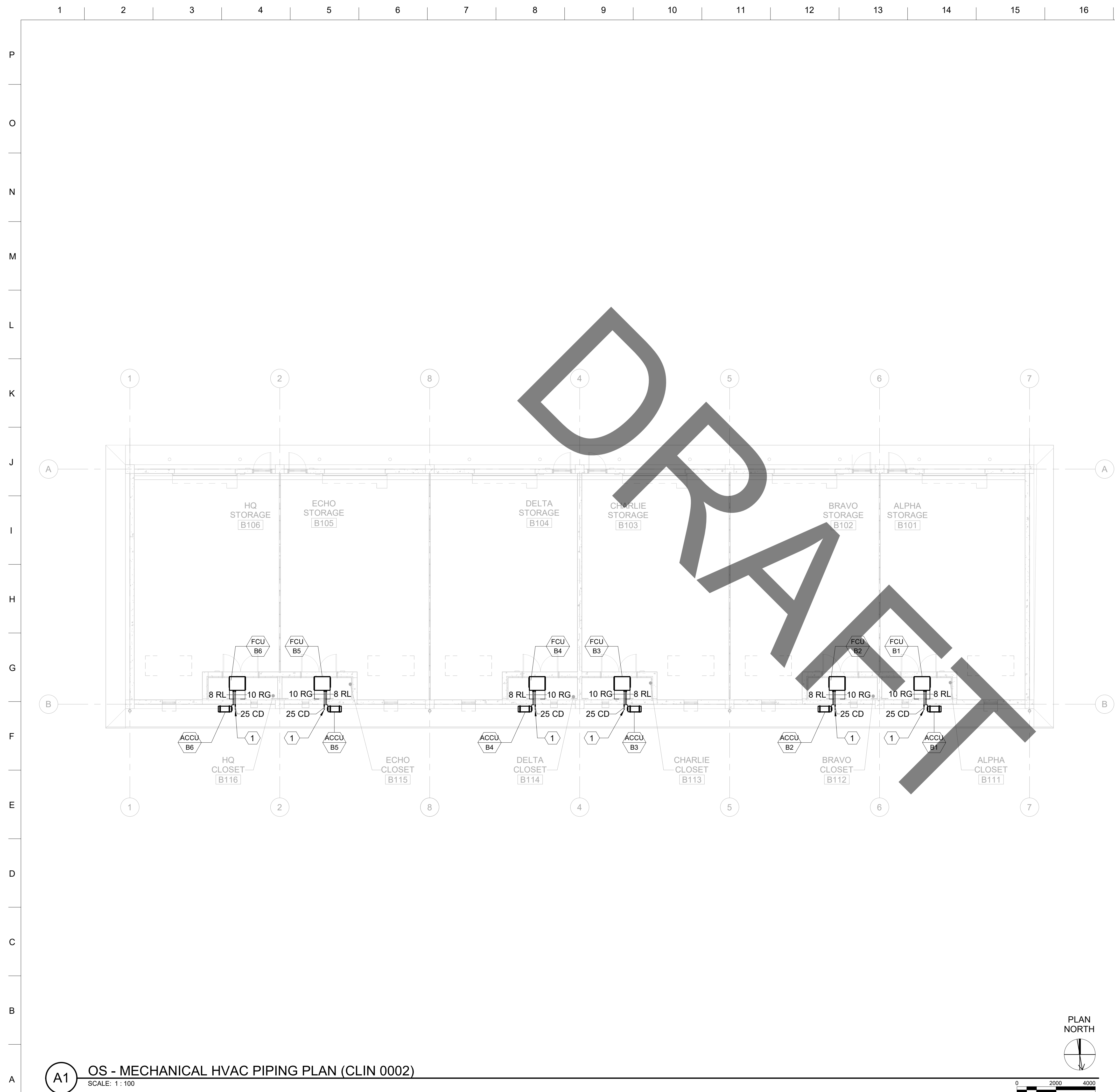
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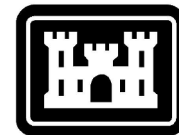


US Army Corps of Engineers®			
OS - MECHANICAL HVAC PLAN (CLIN 0002)		SHEET ID	
APO AP 96349-5010 FY20 MCA PN77833 VEHICLE MAINTENANCE SHOP		M-B101	
US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96376-5132		DESIGNED BY: L.TSANG L.TSANG	
DESIGN PARTNERS INCORPORATED 1580 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814		ISSUE DATE: SOLICITATION NO.: CONTRACT NO.: W912HV-17-D-0003	
CHECKED BY: O. CRABTREE O. CRABTREE		MARK	
SIZE: ANS/D		DESCRIPTION	
DATE		DATE	



KEY NOTES

1 CONDENSATE DISCHARGE TO SPLASHBLOCK.



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[illegible]

US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96370-5132	DESIGNED BY: DRAWN BY: Author	ISSUE DATE: SOLICITATION NO.: CONTRACT NO.: W912KNV-17-D-0003
DESIGN PARTNERS INCORPORATED 1580 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814	CHECKED BY: SUBMITTED BY: SIZE:	

APO AP 96343-5010
FY20 MCA PN77833 VEHICLE MAINTENANCE SHOP

OS - MECHANICAL HVAC PIPING PLAN (CLIN 0002)

OS - MECHANICAL HVAC PIPING PLAN (CLIN 0002)

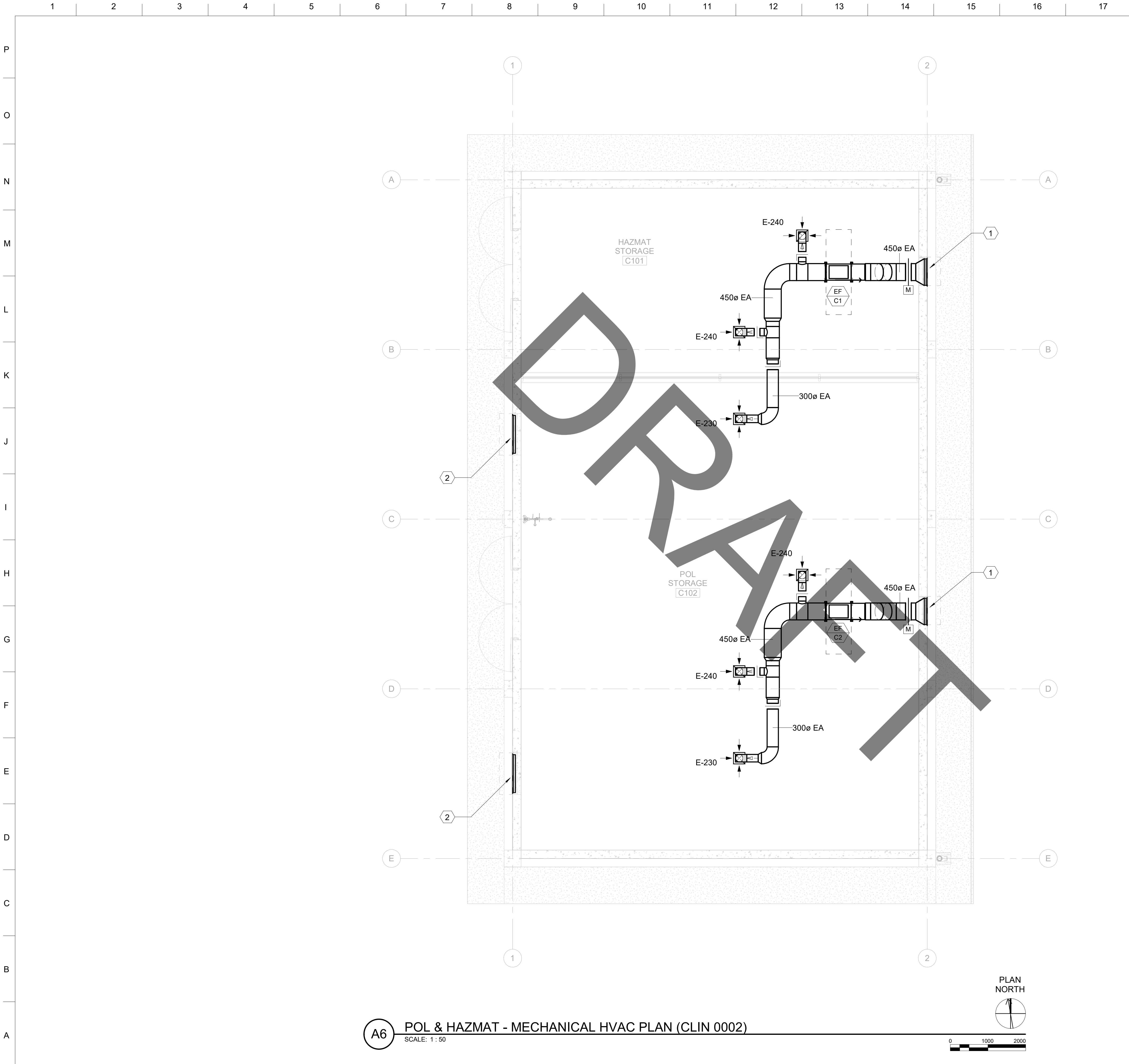
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M-B102

INTERMEDIATE DESIGN

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A6 POL & HAZMAT - MECHANICAL HVAC PLAN (CLIN 0002)
SCALE: 1 : 50

KEY NOTES

- 1 PROVIDE 705 L/S EXHAUST LOUVER WITH MINIMUM 0.19 SQ.M OF FREE AREA.
- 2 PROVIDE 705 L/S INTAKE LOUVER WITH MINIMUM 0.35 SQ.M OF FREE AREA.



MARK	DESCRIPTION	DATE

DESIGNED BY:	ISSUE DATE:
I. TSANG	
DRAWN BY:	SOLICITATION NO.:
R. BALIGANASAY	
CHECKED BY:	CONTRACT NO.:
	W72FW-17-D-0003
SUBMITTED BY:	
O. CRABTREE	
SIZE:	
ANSI D	

US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96376-5132	DESIGN PARTNERS INCORPORATED 1580 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814
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APO AP 96343-5010
FY20 MCA PN77833 VEHICLE MAINTENANCE SHOP

POL & HAZMAT E - MECHANICAL HVAC PLAN (CLIN 0002)

SHEET ID

M-C101

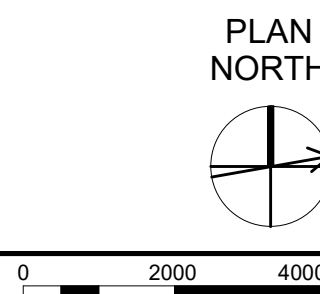
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INTERMEDIATE DESIGN

M-D101

Plot



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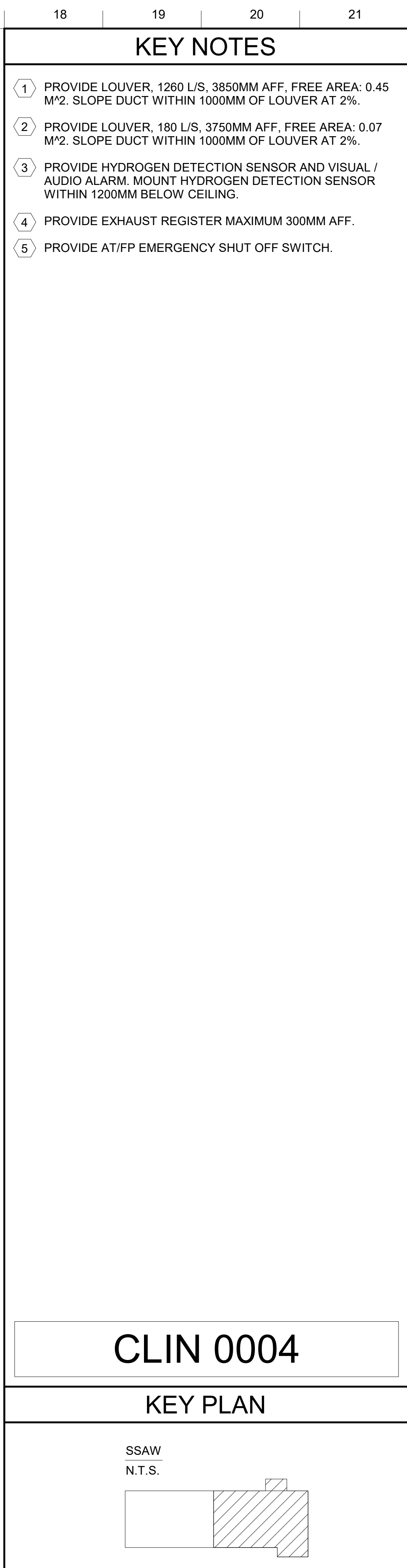
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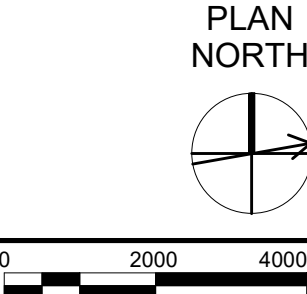
INTERMEDIATE DESIGN

M-D111

INTERMEDIATE DESIGN



APO AP 96345-5010 FY20 MCA PN77833 VEHICLE MAINTENANCE SHOP		SSAAW - PARTIAL MECHANICAL HVAC PLAN - ZONE B (CLIN 0004)	
SHEET ID		M-D112	
US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96376-5132		DESIGNED BY: C. LI	
DESIGN PARTNERS INCORPORATED 1380 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814		ISSUE DATE:	
DRAWN BY: J. K. SAY		SOLICITATION NO.:	
CHECKED BY: O. GRABTREE		CONTRACT NO.: W912HV-17-D-0003	
SUBMITTED BY:		MARK	
SIZE: A3/D		DATE	



- 1 32MM CD TO TERMINATE AT SANITARY STANDPIPE. REFER TO PLUMBING PLANS.
- 2 PROVIDE DDC PANEL.

[illegible]

US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96376-5132	DESIGNED BY:	ISSUE DATE:
	C. LI DRAWN BY: J. KIMURA CHECKED BY: O. CRABTREE SUBMITTED BY:	SOLICITATION NO.: CONTRACT NO.: W912HV-17-D-0003
DESIGN PARTNERS INCORPORATED 1560 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814	SIZE: ANS/D	

SSAW - PARTIAL MECHANICAL HVAC PIPING
PLAN - ZONE B (CLIN 0004)

SHEET ID

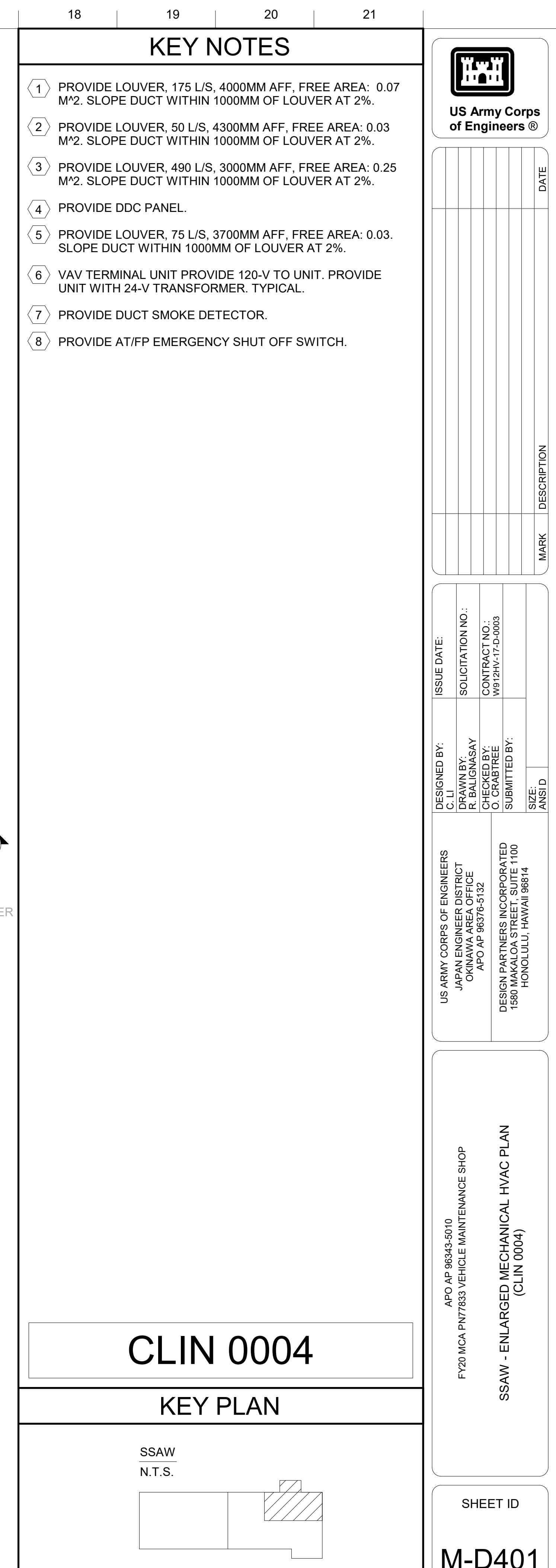
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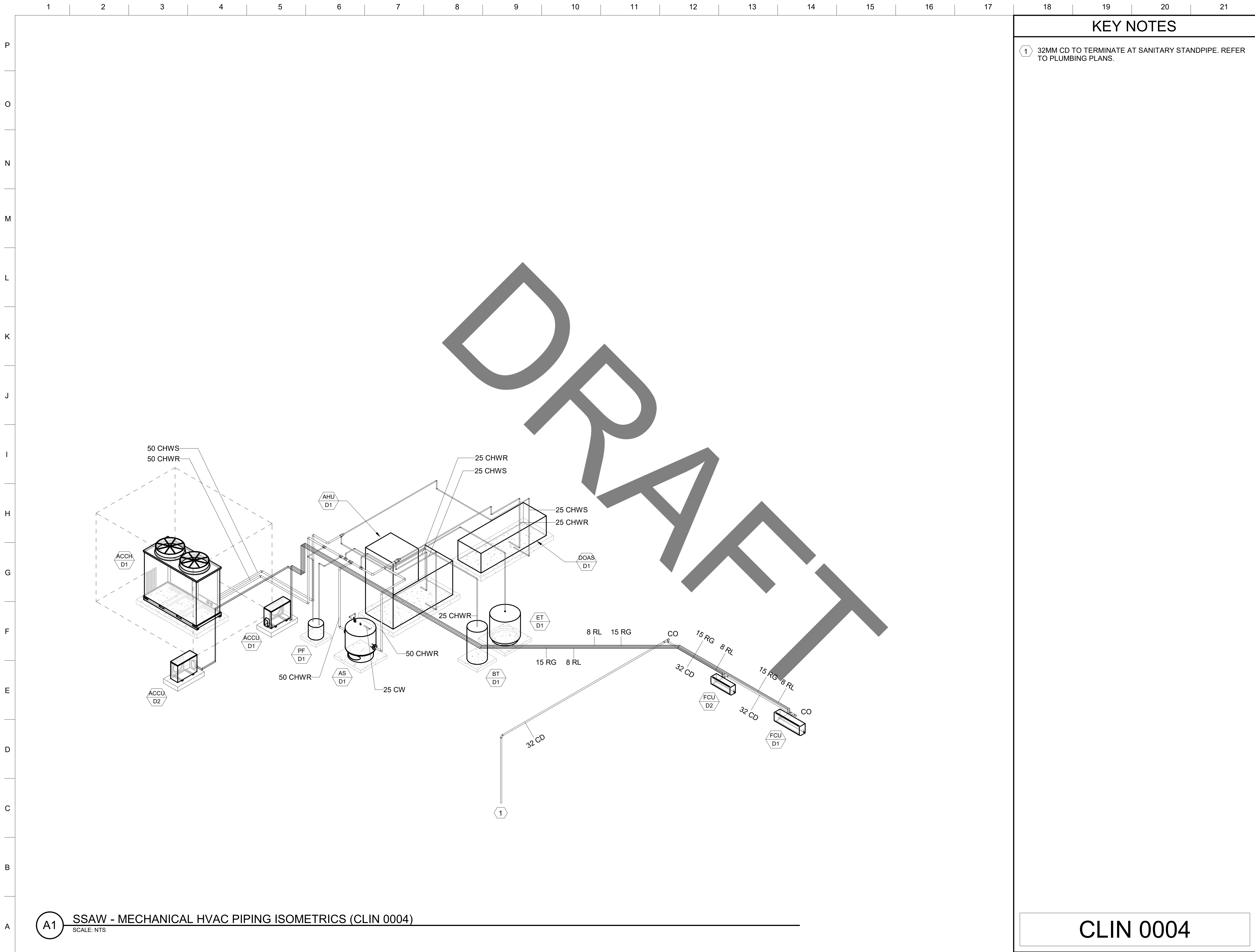
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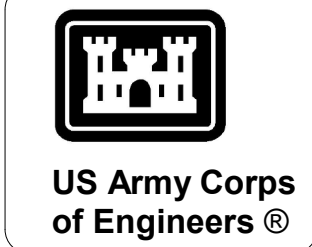






KEY NOTES

1 32MM CD TO TERMINATE AT SANITARY STANDPIPE. REFER TO PLUMBING PLANS.



MARK	DESCRIPTION	DATE

DESIGNED BY: C. LI	ISSUE DATE:
DRAWN BY: R. BALIGASAY	SOLICITATION NO.:
CHECKED BY: W. J. J. J.	CONTRACT NO.:
SUBMITTED BY:	W72HY-17-D-0003
SIZE: ANSI D	

US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96376-5132	
DESIGN PARTNERS INCORPORATED 1580 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814	

APO AP 96343-5010 FY20 MCA PN77833 VEHICLE MAINTENANCE SHOP	
SSAW - MECHANICAL HVAC PIPING ISOMETRICS (CLIN 0004)	

SHEET ID
M-D901

CLIN 0004

[illegible]

KEY NOTES

- 1 EA LOUVER, 230 LPS, MIN. 0.06 SM FREE AREA. SEE ARCH DWGS.
- 2 OA LOUVER, 230 LPS, MIN. 0.11 SM FREE AREA. SEE ARCH DWGS.



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[illegible]

US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT KINAWA AREA OFFICE APO AP 96376-5132	DESIGNED BY: T. TSANG R. BALUNGASAY	ISSUE DATE: SOLICITATION NO.: CONTRACT NO.: W9124H-17-D-0003
DESIGN PARTNERS INCORPORATED 1580 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814	CHECKED BY: O. CRABTREE SUBMITTED BY: O. CRAB TREE	
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APO AP 96343-5010
FY20 MCA PN7833 VEHICLE MAINTENANCE SHOP
FP - MECHANICAL FLOOR PLAN (CLIN 0001)

SHEET ID

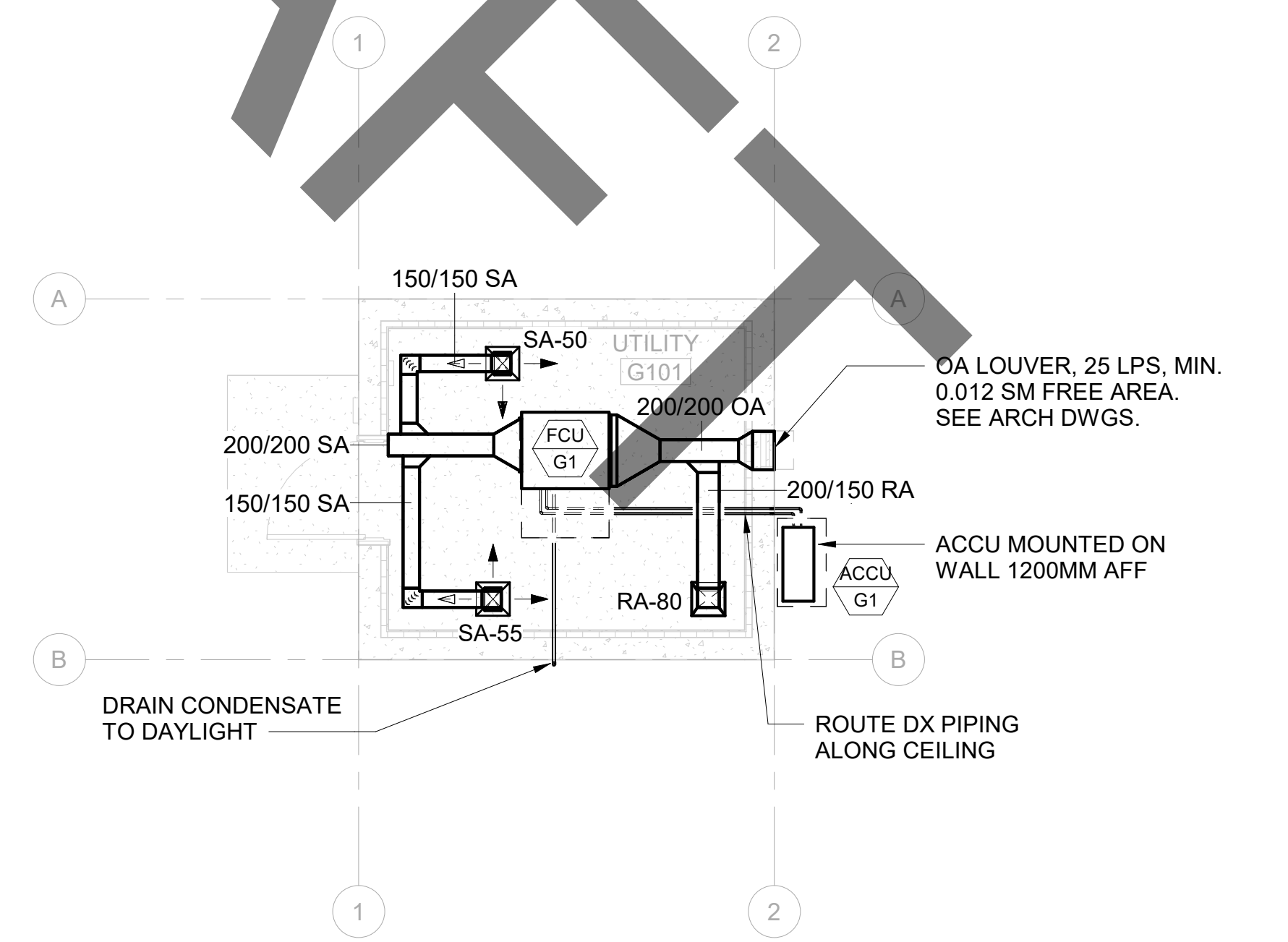
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CLIN 0001

INTERMEDIATE DESIGN

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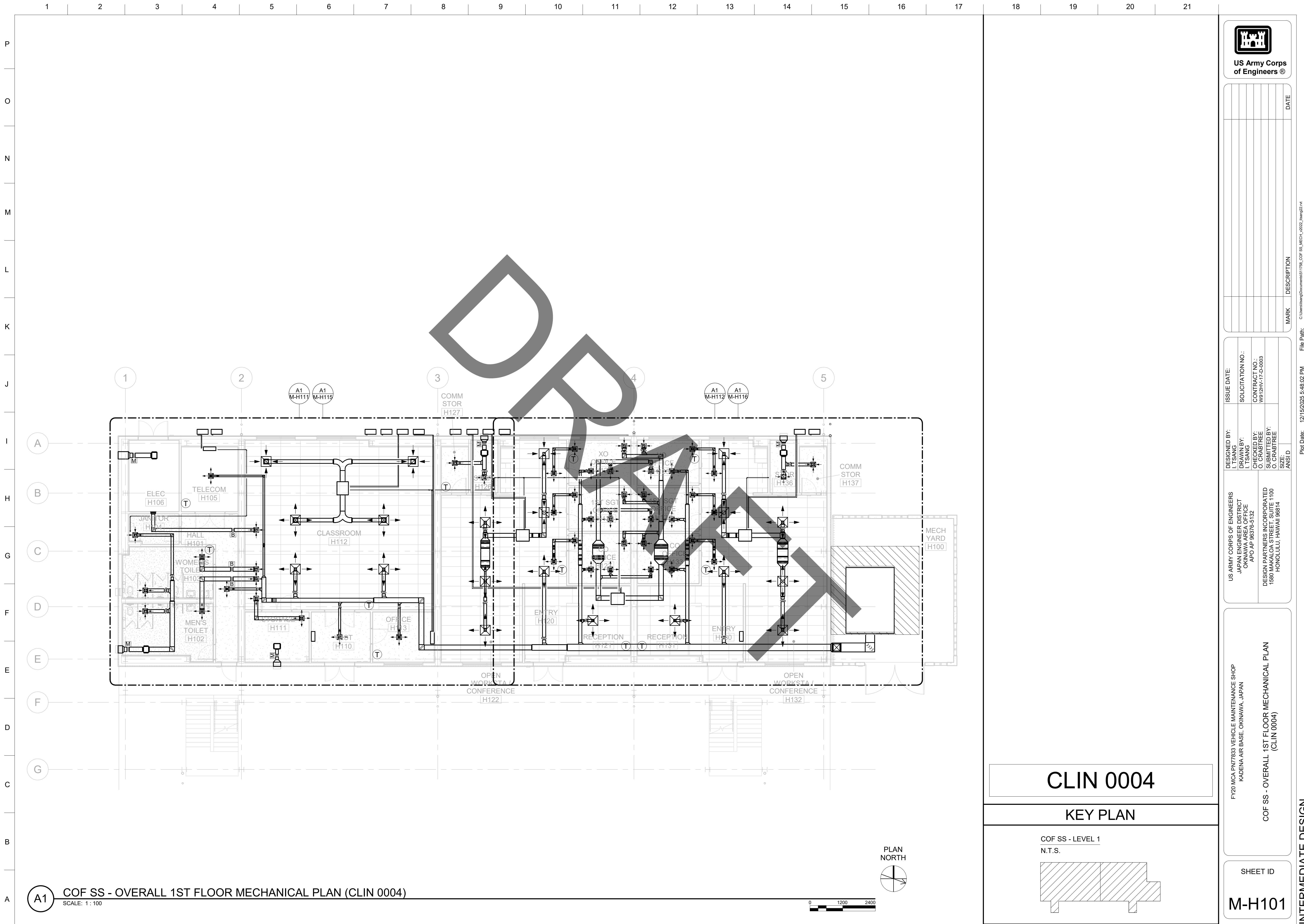
Full Manuscript Ref ID: A111759



CH - MECHANICAL FLOOR PLAN (CLIN 0001)

CLIN 0001

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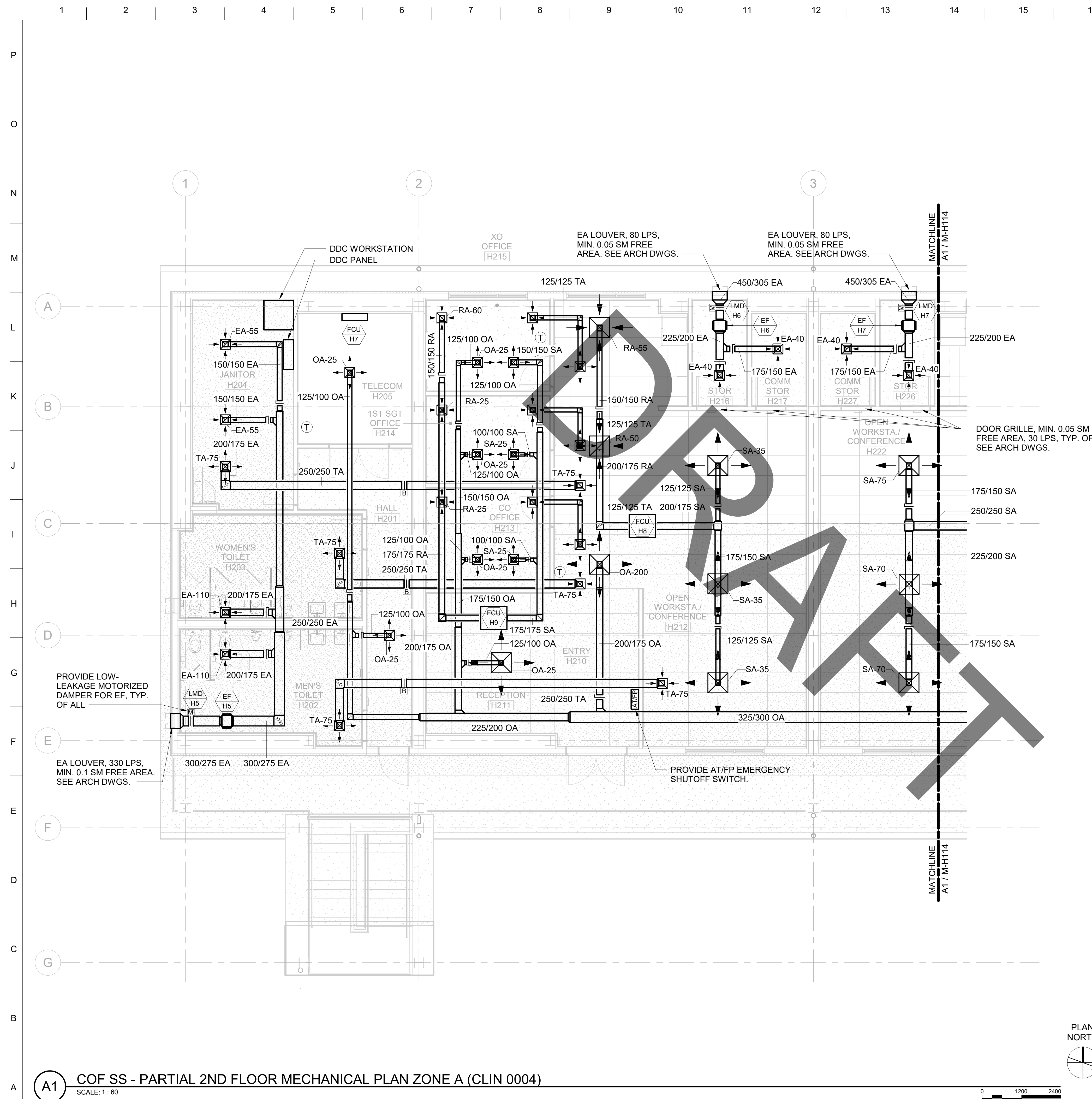


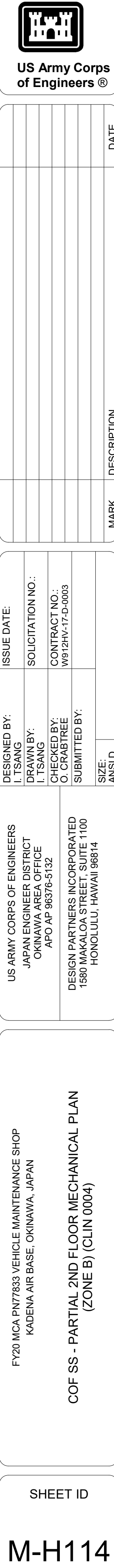


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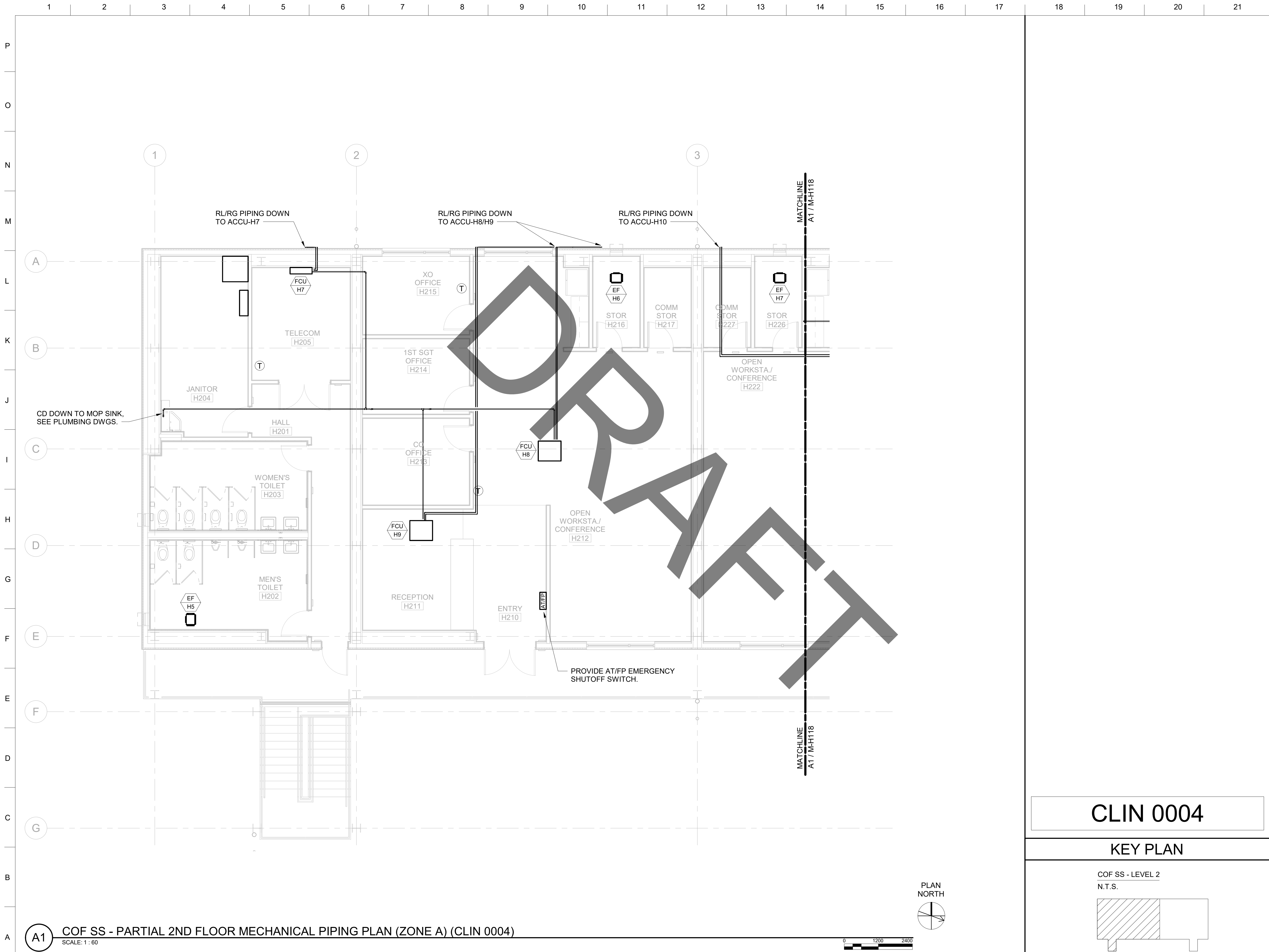










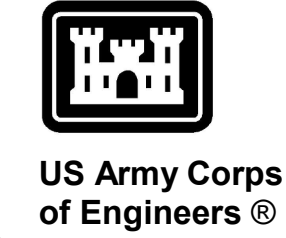
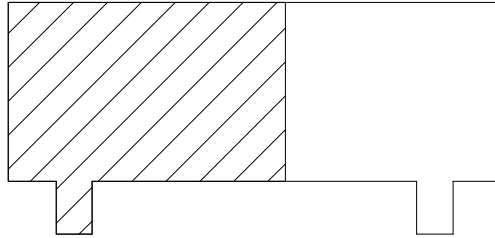


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SCALE: 1:60

CLIN 0004

KEY PLAN

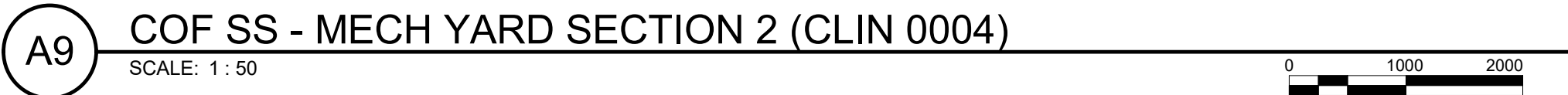
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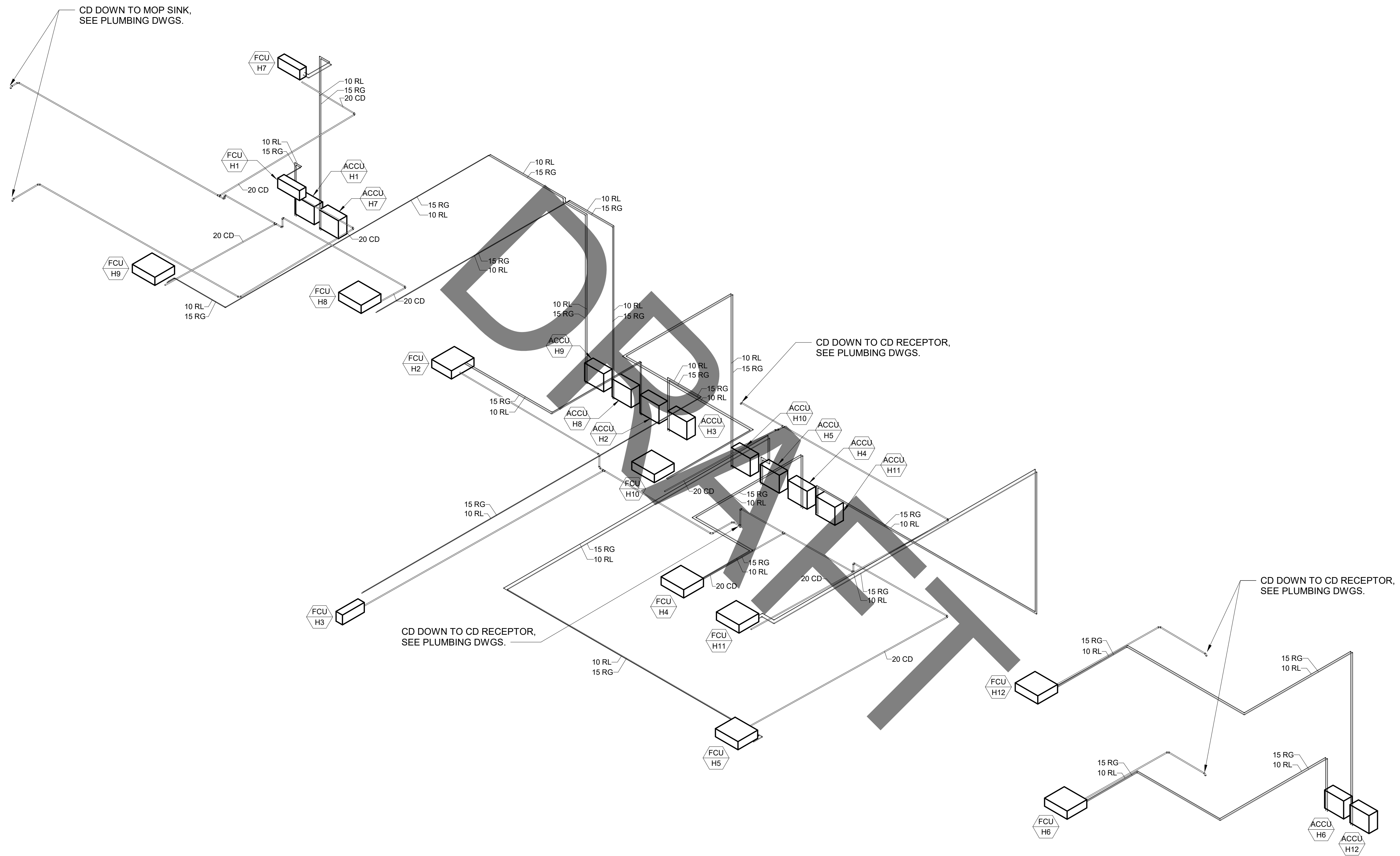
DESIGNED BY: I. TSANG	ISSUE DATE:
DRAWN BY: I. TSANG	SOLICITATION NO.:
CHECKED BY: I. TSANG	CONTRACT NO.:
SUBMITTED BY:	W72WY-17-D-0003
SIZE: ANSI D	

US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA AREA OFFICE
APO AF 96376-5132
DESIGN PARTNERS INCORPORATED
1580 MAKALOA STREET, SUITE 1100
HONOLULU, HAWAII 96814

SHEET ID
M-H117



US Army Corps of Engineers 		DESIGNED BY: I. TSANG		ISSUE DATE:	
DRAWN BY: TSANG		SOLICITATION NO.:		DATE:	
CHECKED BY: O. CRABTREE		CONTRACT NO.: W912HV-17-D-0003		MARK:	
SUBMITTED BY:		SIZE: ANSI/D		DESCRIPTION:	
US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96376-5132					
DESIGN PARTNERS INCORPORATED 1560 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814					



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U.S. ARMY CORPS OF ENGINEERS JANUARY 1991 OKINAWA AREA OFFICE APO AF 96376-5132	DESIGN PARTNERS INCORPORATED 1580 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814	I. TSANG DRAWN BY: CHECKED BY: O' CREAMTREE SUBMITTED BY: DATE: ANSD	SOLICITATION NO.: CONTRACT NO.: W912HJV-17-D-0003
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COF SS - MECHANICAL HVAC PIPING ISOMETRICS
(CLIN 0004)

SHEET ID

M-H901

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INTERMEDIATE DESIGN

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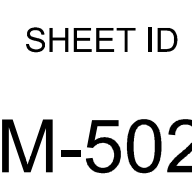
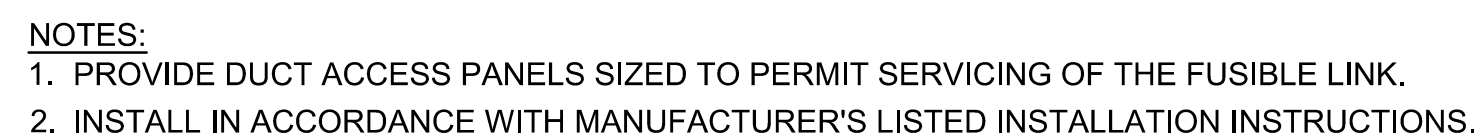
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A1

COF SS - MECHANICAL HVAC PIPING ISOMETRICS (CLIN 0004)

SCALE: NTS

CLIN 0004



US Army Corps of Engineers®

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CH DIRECT EXPANSION ACCU SCHEDULE (CLIN 0001)

TAG	BUILDING LOCATED AT	AREA SERVED	TOTAL COOLING CAPACITY (KW)	AMBIENT AIR TEMP (C)	POWER (KW)	FLA (A)	MOCP (A)	V/PH/HZ	SOUND (DBA)	DIMENSIONS HxWxD (MM)	WEIGHT (KG)	REMARKS
ACCU-G1	CABLE HUT	G101 UTILITY	2.6	35	1.61	8.56	15	200-208/1/60	61	550X675X313	28.6	1

NOTES:
1. PROVIDE WITH CORROSION PROTECTION, MINIMUM RATING 6000 HOURS OF ASTM B117-97 SALT SPRAY TEST.

VMS EF SCHEDULE (CLIN 0001)

TAG	BUILDING LOCATED AT	AREA SERVED	TYPE	DRIVE	L/S	ESP (PA)	FAN RPM	MOTOR SIZE (kW)	FLA	V/PH/Hz	OPERATING WEIGHT (KG)	NOISE LEVEL (DBA)	REMARKS
EF-A1.1	VMS	A108 MEN/A109 WOMEN/A110 JAN	CENTRIFUGAL INLINE FAN	DIRECT	425	184.2	1070	0.15	1.9	100/1/60	52	-	1,2,10
EF-A1.2	VMS	A111 NON SENSITIVE SECURE STORAGE	CENTRIFUGAL INLINE FAN	DIRECT	215	148.8	1360	0.04	0.84	100/1/60	22	-	1,2,5
EF-A1.3	VMS	A115 FLUID DISTRIBUTION ROOM	CENTRIFUGAL INLINE FAN	BELT	380	153.2	1070	0.15	1.9	100/1/60	52	-	1,2,4,6
EF-A1.4	VMS	VEHICLE MAINTENANCE SHOP, REPAIR BAYS, MACHINE SHOP/WELDING AREA, COMPONENT WASH BAY, CENTRAL VEHICLE CORRIDOR/MAINTENANCE AREA, MAINTENANCE PIT	CENTRIFUGAL UTILITY FAN	BELT	12,965	512.9	-	18.5	-	200-208/3/60	440	-	1,2,3,5,6,7,8
EF-A1.5	VMS	A101B REPAIR BAYS	CENTRIFUGAL UTILITY FAN WITH TAIL PIPE EXHAUST	BELT	660	435.2	-	1.5	-	200-208/3/60	23	-	1,2,3,5,6,11
EF-A1.6	VMS	A101B REPAIR BAYS	CENTRIFUGAL UTILITY FAN WITH TAIL PIPE EXHAUST	BELT	660	435.2	-	1.5	-	200-208/3/60		-	1,2,3,5,6,11
EF-A1.7	VMS	A101B REPAIR BAYS	CENTRIFUGAL UTILITY FAN WITH TAIL PIPE EXHAUST	BELT	660	435.2	-	1.5	-	200-208/3/60	130	-	1,2,3,5,6,11
EF-A1.8	VMS	A101C REPAIR BAYS	CENTRIFUGAL UTILITY FAN WITH TAIL PIPE EXHAUST	BELT	660	435.2	-	1.5	-	200-208/3/60	130	-	1,2,3,5,6,11
EF-A1.9	VMS	A101C REPAIR BAYS	CENTRIFUGAL UTILITY FAN WITH TAIL PIPE EXHAUST	BELT	660	435.2	-	1.5	-	200-208/3/60	130	-	1,2,3,5,6,11
EF-A1.10	VMS	A101C REPAIR BAYS	CENTRIFUGAL UTILITY FAN WITH TAIL PIPE EXHAUST	BELT	660	435.2	-	1.5	-	200-208/3/60	130	-	1,2,3,5,6,11
EF-A1.11	VMS	A102 CENTRAL VEHICLE CORRIDOR/MAINTENANCE AREA	CENTRIFUGAL UTILITY FAN WITH TAIL PIPE EXHAUST	BELT	660	196.7	-	1.5	-	200-208/3/60	130	-	1,2,3,5,6,11
EF-A1.12	VMS	A102 CENTRAL VEHICLE CORRIDOR/MAINTENANCE AREA	CENTRIFUGAL UTILITY FAN WITH TAIL PIPE EXHAUST	BELT	660	210.7	-	1.5	-	200-208/3/60	130	-	1,2,3,5,6,11
EF-A1.13	VMS	A114 ARMS VAULT	CENTRIFUGAL INLINE FAN	DIRECT	55	126.4	1735	0.012	0.42	100/1/60	11	-	1,2,3,5,6,11
EF-A2.1	VMS	A208 RECYCLING	CENTRIFUGAL INLINE FAN	DIRECT	60	137.0	1735	0.012	0.42	100/1/60	11	-	1,2,5
EF-A2.2	VMS	A210 WOMEN/A211 MEN/A212 JAN	CENTRIFUGAL INLINE FAN	DIRECT	195	154.4	1360	0.04	0.84	100/1/60	22	-	1,2,10

NOTES:
1. PROVIDE WITH INTEGRAL BACKDRAFT DAMPER.
2. PROVIDE INTEGRAL DISCONNECT SWITCH.
3. PROVIDE WITH VFD.
4. FAN SHALL OPERATE ON A CONTINUOUS BASIS.
5. FAN SHALL OPERATE CONTINUOUSLY DURING BUSINESS HOURS.
6. FAN SHALL BE OF TYPE A SPARK RESISTANT CONSTRUCTION. FAN MOTOR SHALL NOT BE LOCATED IN AIR STREAM AND SHALL BE IN AN EXPLOSION PROOF ENCLOSURE.
7. INTERLOCK WITH CO SENSOR.
8. INTERLOCK WITH NOX SENSOR.
9. INTERLOCK WITH FORKLIFT CHARGER(S).
10. INTERLOCK WITH SPACE OCCUPANCY SENSORS.
11. PROVIDE WITH HOSE REEL AND TAILPIECE EXHAUST ACCESSORY.
12. INTERLOCK WITH THERMOSTAT.

FPMP EF SCHEDULE (CLIN 0001)

TAG	BUILDING LOCATED AT	AREA SERVED	TYPE	DRIVE	L/S	ESP (PA)	FAN RPM	MOTOR SIZE (kW)	FLA	V/PH/Hz	OPERATING WEIGHT (KG)	NOISE LEVEL (DBA)	REMARKS
EF-F1	FIRE PUMP	F101 FIRE PUMP	CENTRIFUGAL INLINE FAN	DIRECT	230	62.2	1360	0.05	0.84	100/1/60	39	-	1,2,3

NOTES:
1. PROVIDE WITH INTEGRAL BACKDRAFT DAMPER.
2. PROVIDE INTEGRAL DISCONNECT SWITCH.
3. INTERLOCK WITH THERMOSTAT.

VMS MAF SCHEDULE (CLIN 0001)

TAG	BUILDING LOCATED AT	AREA SERVED	TYPE	DRIVE	L/S	ESP (PA)	FAN RPM	MOTOR SIZE (HP)	FLA	V/PH/Hz	OPERATING WEIGHT (KG)	NOISE LEVEL (DBA)	REMARKS
MAF-A1	VMS	VEHICLE MAINTENANCE SHOP, REPAIR BAYS, MACHINE SHOP/WELDING AREA, COMPONENT WASH BAY, CENTRAL VEHICLE CORRIDOR/MAINTENANCE AREA, MAINTENANCE PIT	CENTRIFUGAL UTILITY FAN	BELT	16,815	373.3	615	20	59.4	200-208/3/60	949	80	1,2,3,4

NOTES:
1. PROVIDE WITH INTEGRAL BACKDRAFT DAMPER.
2. PROVIDE INTEGRAL DISCONNECT SWITCH.
3. PROVIDE WITH VFD.
4. INTERLOCK FAN WITH EF-A1.6 AND EF-A1.7.

VMS VHR SCHEDULE (CLIN 0001)

UNIT	LOCATION	SYSTEM SERVED	AIRFLOW (LPS)	HOSE DIAMETER (MM)	HOSE LENGTH (M)	REMARKS
VHR-A1.1	REPAIR BAYS	EF-1.5	660	250	13.4	1,2,3,4,5,6
VHR-A1.2	REPAIR BAYS	EF-1.6	660	250	13.4	1,2,3,4,5,6
VHR-A1.3	REPAIR BAYS	EF-1.7	660	250	13.4	1,2,3,4,5,6
VHR-A1.4	REPAIR BAYS	EF-1.8	660	250	13.4	1,2,3,4,5,6
VHR-A1.5	REPAIR BAYS	EF-1.9	660	250	13.4	1,2,3,4,5,6
VHR-A1.6	REPAIR BAYS	EF-1.10	660	250	13.4	1,2,3,4,5,6
VHR-A1.7	REPAIR BAYS	EF-1.11	660	250	13.4	1,2,3,4,5,6
VHR-A1.8	REPAIR BAYS	EF-1.12	660	250	13.4	1,2,3,4,5,6

1. PROVIDE ALL MOUNTING HARDWARE AND STRUCTURAL SUPPORT.
2. COORDINATE HOSE REEL LOCATION AND HEIGHT WITH VEHICLE BAYS AND CRANE RAILS.
3. PROVIDE POWER INTERLOCK WHERE SPECIFIED FOR AUTOMATIC FAN START.
4. ALL EXHAUST HOSES TO BE HIGH-TEMPERATURE NEOPRENE FIBERGLASS OR EQUIVALENT.
5. CONNECTION TO EXHAUST FAN DUCTWORK BY CONTRACTOR; VERIFY DUCT SIZE AND TRANSITIONS.
6. PROVIDE TAILPIPE ADAPTERS AS SCHEDULED OR REQUIRED IN FIELD.

VMS LMD SCHEDULE (CLIN 0001)

TAG	SERVES	V	PH	HZ	W	REMARKS
LMD-A1	EF-A1.1	24	1	60	5.5	1,2
LMD-A2	EF-A1.2	24	1	60	5.5	1,2
LMD-A3	EF-A1.3	24	1	60	5.5	1,2
LMD-A4	EF-A1.4	24	1	60	5.5	1,2
LMD-A5	EF-A1.13	24	1	60	5.5	1,2
LMD-A6	EF-A2.1	24	1	60	5.5	1,2
LMD-A7	EF-A2.2	24	1	60	5.5	1,2
LMD-A8	DOAS-A1	24	1	60	5.5	1,2

1. PROVIDE SIZE AS INDICATED ON DRAWINGS TO COINCIDE W/ DUCT.
2. PROVIDE INTEGRAL TRANSFORMER

US Army Corps of Engineers®

DESIGNED BY:
L. TSANG

DRAWN BY:
S. KIM

CHECKED BY:
O. USABREE

SUBMITTED BY:
O. CRABTREE

SIZE:
ANSI D

ISSUE DATE:

SOLICITATION NO.:

CONTRACT NO.:

SCALE:

AS SHOWN

US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA AREA OFFICE
APO AP 96376-5132

DESIGN PARTNERS INCORPORATED
1580 MAKALOA STREET, SUITE 1100
HONOLULU, HAWAII 96814

APO AP 96343-5010
FY20 MCA PN77833 VEHICLE MAINTENANCE SHOP
MECHANICAL SCHEDULES

SHEET ID

M-602

INTERMEDIATE DESIGN

VALVE NO.	EQUIPMENT	CAPACITY (LPM)	MAX PRESSURE DROP	VALVE SIZE (MM)	VALVE CV AT FULL FLOW	VALVE TYPE	CONFIGURATION	FAIL POSITION	MIN. CLOSOFF PRESS. (KPA)	MATERIAL
			KPA							
CV-AHU-A1	AHU-A1	95.8	35	40	18.3	BALL	2-WAY	IN PLACE	690	NICKEL-PLATED BRASS BALL VALVE, TEFLON STEALS, ANGLE ROTATION: 0-90 DEG
CV-AHU-A2	AHU-A2	90.8	35	40	15.1	BALL	2-WAY	IN PLACE	690	
CV-DOAS-A1	DOAS-A1	179.0	35	40	32.4	BALL	2-WAY	IN PLACE	690	

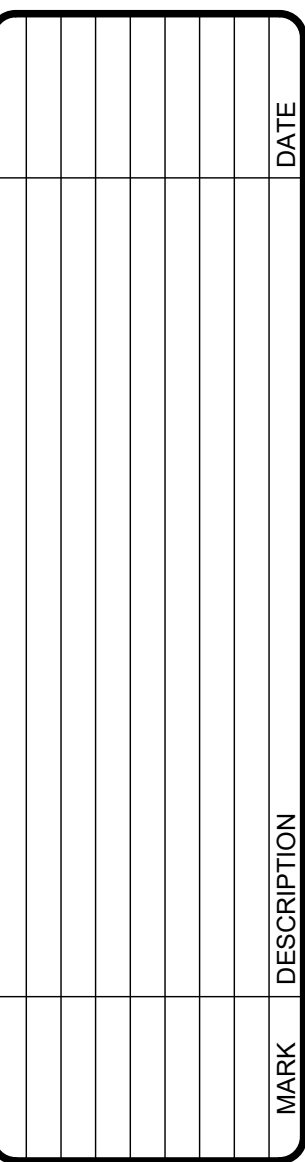
SYSTEM	DUCT PRESSURE CLASS (PA)				SUPPLY				RETURN AIR/OUTSIDE AIR		DUCT PRESSURE TEST MM OF WATER COLUMN	NOTES
					ROUND/OVAL		RECTANGULAR		DUCT SEAL CLASS	DUCT LEAK CLASS		
	SUPPLY DUCT	RETURN DUCT	EXHAUST DUCT	OUTSIDE AIR DUCT	DUCT SEAL CLASS	DUCT LEAK CLASS	DUCT SEAL CLASS	DUCT LEAK CLASS				
AHU-A1	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
AHU-A2	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
DOAS-A1	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-	-	500	-	-	-	-	A	24	50	[1]
FCU-A4	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-	-	500	-	-	-	-	A	24	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
FCU-A5	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-	-	500	-	-	-	-	A	24	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
EF-A1.1	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-A1.2	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-A1.3	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-A1.4	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-A1.5	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-A1.6	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-A1.7	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-A1.8	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-A1.9	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-A1.10	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-A1.11	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-A1.12	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-A1.13	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-A2.1	-	-	-250	-	A	12	A	24	-	-	25	[1]

VAV TAG	ROOM	NAME	AREA (M^2)	MIN SUPPLY AIR (L/s)	DESIGN SUPPLY AIR (L/s)	DESIGN RETURN AIR (L/s)	DESIGN OA (L/s)	UNCONDITIONED OA (L/s)	DESIGN EXHAUST AIR(L/s)	TRANSFER IN (L/s)	TRANSFER OUT (L/s)	BALANCE (L/s)	SERVED BY	REMARKS
1ST FLOOR														
VAV-A103	A103	TOOL ROOM	78	0	285	285	40	0	0	0	0	40	AHU-A1 / DOAS-A1	
VAV-A104	A104	COMBAT SPARES	91	0	330	330	45	0	0	0	0	45	AHU-A1 / DOAS-A1	
VAV-A105	A105	ELECTRICAL ROOM	30	0	950	950	0	0	0	0	0	0	AHU-A1 / DOAS-A1	
VAV-A107	A107	MECHANICAL ROOM	107	0	630	630	0	0	0	0	0	0	AHU-A1 / DOAS-A1	
-	A108	MEN	43	0	0	0	0	230	230	0	0	0	EF-A1.1	
-	A109	WOMEN	27	0	0	0	0	165	165	0	0	0	EF-A1.1	
-	A110	JAN	5	0	0	0	0	30	30	0	0	0	EF-A1.1	
VAV-A112	A112	CONSOLIDATED BENCH REPAIR	116	0	435	435	60	0	0	0	0	60	AHU-A1 / DOAS-A1	
VAV-A116	A116	COMPRESSOR ROOM	23	0	165	165	0	0	0	0	0	0	AHU-A1 / DOAS-A1	
-	A117	ELEVATOR EQUIPMENT ROOM	8	0	300	300	0	0	0	0	0	0	AHU-A1 / DOAS-A1	
VAV-A140	A140	NURSING ROOM	5	0	25	25	25	0	0	0	0	25	AHU-A1 / DOAS-A1	
TOTAL (SUM OF PEAKS)						3120	3120	170	425	425	0	0	170	
2ND FLOOR														
VAV-A201	A201	CORRIDOR/ELEV LOBBY	56	0	25	25	25	0	0	0	0	25	AHU-A2 / DOAS-A1	
VAV-A202	A202	ADMIN & SHOP CONTROL	167	0	525	525	85	0	0	0	0	85	AHU-A2 / DOAS-A1	
VAV-A203	A203	OFFICE 1	10	0	30	30	25	0	0	0	0	25	AHU-A2 / DOAS-A1	
VAV-A204	A204	OFFICE 2	10	0	30	30	25	0	0	0	0	25	AHU-A2 / DOAS-A1	
VAV-A205	A205	TRAINING ROOM	99	0	540	540	160	0	0	0	0	160	AHU-A2 / DOAS-A1	
VAV-A206	A206	ADMIN & SHOP CONTROL OVERFLOW	87	0	455	455	45	0	0	0	0	45	AHU-A2 / DOAS-A1	
-	A208	RECYCLING	11	0	0	0	0	60	60	0	0	0	EF-A1.2.1	
VAV-A209	A209	BREAK/TRAINING/CONFERENCE	61	0	360	360	110	0	0	0	0	110	AHU-A2 / DOAS-A1	
-	A210	WOMEN	24	0	0	0	0	75	75	0	0	0	EF-A2.2	
-	A211	MEN	23	0	0	0	0	95	95	0	0	0	EF-A2.2	
-	A212	JAN	3	0	0	0	0	25	25	0	0	0	EF-A2.2	
VAV-A214	A214	ELEC	4	0	825	825	0	0	0	0	0	0	AHU-A2 / DOAS-A1	
TOTAL (SUM OF PEAKS)						2790	2790	475	255	255	0	0		
BUILDING PRESSURIZATION (%)													5%	

[illegible]

EQUIPMENT	BASE TYPE	TYPE OF ISOLATOR	APPLICATION	DEFLECTION (MM)
AHU-A1	A	3	PACKAGED AH, AC, H, AND V UNITS	38
AHU-A2	A	3	PACKAGED AH, AC, H, AND V UNITS	38
DOAS-A1	A	3	PACKAGED AH, AC, H, AND V UNITS	38
ACCH-A1	A	4	REFRIGERANT MACHINES AND CHILLERS	19
FCU-A4	A	3	PACKAGED AH, AC, H, AND V UNITS	38
FCU-A5	A	3	PACKAGED AH, AC, H, AND V UNITS	38
ACCU-A1	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-A2	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-A3	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-A4	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-A5	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-A6	A	1	REFRIGERANT MACHINES AND CHILLERS	19
EF-A1.1	A	3	EXHAUST FAN	38
EF-A1.2	A	3	EXHAUST FAN	38
EF-A1.3	A	3	EXHAUST FAN	38
EF-A1.4	A	3	EXHAUST FAN	38
EF-A1.5	A	3	EXHAUST FAN	38
EF-A1.6	A	3	EXHAUST FAN	38
EF-A1.7	A	3	EXHAUST FAN	38
EF-A1.8	A	3	EXHAUST FAN	38
EF-A1.9	A	3	EXHAUST FAN	38
EF-A1.10	A	3	EXHAUST FAN	38
EF-A1.11	A	3	EXHAUST FAN	38
EF-A1.12	A	3	EXHAUST FAN	38
EF-A1.13	A	3	EXHAUST FAN	38
EF-A2.1	A	3	EXHAUST FAN	38
EF-A2.2	A	3	EXHAUST FAN	38

ISOLOATOR TYPE:
TYPE 1: PAD, RUBBER, OR GLASS FIBER
TYPE 2: RUBBER FLOOR ISOLATOR OR HANGER
TYPE 3: SPRING FLOOR ISOLATOR OR HANGER
TYPE 4: RESTRAINED SPRING ISOLATOR
TYPE 5: THRUST RESTRAINT
TYPE 6: AIR SPRING



DESIGNED BY: US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96376-5152	DRAWN BY: I. TSANG	SOLICITATION NO.:	ISSUE DATE:
DESIGN PARTNERS INCORPORATED 1980 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814	CHECKED BY: C. CRIBBIE	CONTRACT NO.:	
	DATE: 01/01/01	W912HV-17-D-0003	
	DATE: 01/01/01	AS SHOWN	
SIZE:	ANSI D		

APO AP 96343-5010
FY20 MCA PN77833 VEHICLE MAINTENANCE SHOP
MECHANICAL SCHEDULES

INTERMEDIATE DESIGN

M-604

	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21		
P	SSAW CHILLED WATER DOAS SCHEDULE (CLIN 0004)																					
	TAG	BUILDING LOCATED AT	AREA SERVED	TYPE	FAN		COOLING								ELECTRICAL			SIZE LxWxH (MM)	WEIGHT (KG)	REMARKS		
					OUTDOOR AIR FLOW (LPS)	ESP (PA)	TOTAL (KW)	SENS. (KW)	DB (°C)	WB (°C)	DB (°C)	WB (°C)	CHWS (C°)	CHWR (C°)	WPD (M. HD.)	LPM	FLA (A)	POWER (KW)	V/PH/Hz			
	DOAS-D1	SSA WAREHOUSE	OFFICES, BREAKROOM, CONFERENCE ROOM, MECH/ELEC ROOM, CORRIDOR, SECURE STORAGE, TELECOM, SIPR ROOM	HORIZONTAL INLINE	515	340.0	29.2	11.5	31.7	26.6	12.8	12.1	6.7	14.4	1.40	54.5	5.1	2.4	200-208/3/60	2743x965x711	609	1-10
	1. SUPPLY AIR FAN(S) WITH VFD. PROVIDE INTEGRAL DISCONNECT																					
	2. EXTERNAL PRESSURE LOSS EXCLUDES FILTER																					
	3. DOUBLE WALL CONSTRUCTION W/ 2" THICK INSULATION																					
	4. PROVIDE DUCT SMOKE DETECTOR. DUCT SMOKE DETECTOR SHALL BE FURNISHED BY FIRE PROTECTION CONTRACTOR AND INSTALLED BY MECHANICAL. COORDINATE WITH ELECTRICAL																					
	5. COOLING COIL AND FIN SHALL BE ALL COPPER																					
	6. FINISH WITH SECONDARY DRAIN PAN. CONNECT TO SANITARY																					
	7. INTERNAL SPRING ISOLATORS FOR FANS SHALL BE TIED DOWN. COMPLETE SEISMIC RESTRAINED SPRING ISOLATORS, 2" DEFLECTION																					
	8. PROVIDE WITH MERV 8 PREFILTER AND MERV 13 FINAL FILTER																					
M	9. PROVIDE UVC LIGHT. UVC LIGHT SHALL BE INTERLOCKED WITH CABINET DOOR																					
	10. UNIT SHALL HAVE A MINIMUM OF 500 HOUR SALT SPRAY COATED COILS. PROVIDE COILS WITH PHENOLIC COATING FOR CORROSION PROTECTION.																					
L	COF SS PACKAGED DOAS SCHEDULE (CLIN 0004)																					
	TAG	BUILDING LOCATED AT	AREA SERVED	TYPE	FAN		COOLING COIL						HOT GAS REHEAT COIL		ELECTRICAL				SIZE LxWxH (MM)	WEIGHT (KG)	REMARKS	
					OUTDOOR AIR FLOW (LPS)	ESP (PA)	TOTAL (KW)	SENS. (KW)	DB (°C)	WB (°C)	DB (°C)	WB (°C)	TOTAL (KW)	DB (°C)	WB (°C)	MCA (A)	MOCp (A)	FLA	V/PH/Hz			
	DOAS-H1	COF SS	CLASSROOM, OFFICES, OPEN WORKSTA./CONFERENCE, TELECOM	HORIZONTAL	1455	322.1	97.2	38.7	32.0	26.7	12.6	11.9	24.6	23.9	17.0	126	150	115	200-208/3/60	4028x2950x1753	1,316	1-13
	1. SUPPLY AIR FAN(S) WITH VFD. PROVIDE INTEGRAL DISCONNECT WITH A NEMA 4X ENCLOSURE.																					
	2. EXTERNAL PRESSURE LOSS EXCLUDES FILTER																					
	3. DOUBLE WALL CONSTRUCTION W/ 2" THICK INSULATION																					
	4. PROVIDE DUCT SMOKE DETECTOR. DUCT SMOKE DETECTOR SHALL BE FURNISHED BY FIRE PROTECTION CONTRACTOR AND INSTALLED BY MECHANICAL. COORDINATE WITH ELECTRICAL																					
	5. COOLING COIL AND FIN SHALL BE ALL COPPER																					
	6. FINISH WITH SECONDARY DRAIN PAN. CONNECT TO SANITARY																					
	7. PROVIDE WITH INTERNAL SPRING ISOLATORS																					
J	8. PROVIDE WITH MERV 8 PREFILTER AND MERV 13 FINAL FILTER.																					
	9. PROVIDE UVC LIGHT. UVC LIGHT SHALL BE INTERLOCKED WITH CABINET DOOR																					
	10. PROVIDE HOT GAS REHEAT COIL																					
	11. UNIT SHALL HAVE A MINIMUM OF 500 HOUR SALT SPRAY COATED COILS. PROVIDE COILS WITH PHENOLIC COATING FOR CORROSION PROTECTION.																					
	12. UNIT SHALL BE HORIZONTAL CONFIGURATION WITH END SUPPLY, NO RETURN																					
	13. PROVIDE UNIT WITH RAINHOOD AND MOTORIZED DAMPER FOR THE INTAKE SIDE.																					
H	SSAW DIRECT EXPANSION FCU SCHEDULE (CLIN 0004)																					
	TAG	BUILDING LOCATED AT	AREA SERVED	TYPE	FAN			COOLING				ELECTRICAL				MAX NOISE (DBA)	SIZE HxWxD (MM)	WEIGHT (KG)	REMARKS			
					SA (LPS)	OA (LPS)	ESP (PA)	TOTAL (KW)	SENS. (KW)	DB (°C)	WB (°C)	DB (°C)	WB (°C)	POWER (KW)	FLA (A)	MOP (A)	V/PH/Hz					
	FCU-D1	SSA WAREHOUSE	D107 SIPR ROOM	WALL MOUNTED	190	0	N/A	2.3	2.2	22.2	14.9	12.8	10.9	POWERED BY ACCU				55	293x798x230	10.0	1, 2, 3	
	FCU-D2	SSA WAREHOUSE	D109 TELECOM	WALL MOUNTED	395	0	N/A	4.8	4.6	22.2	14.6	12.8	10.6	POWERED BY ACCU				36.7	320x1050x250	13.0	1, 2, 3	
	NOTES: 1. FCU IS POWERED BY ACCU. 2. PROVIDE WITH MERV 8 FILTER. 3. PROVIDE WITH CONDENSATE PUMP.																					
G																						
F	COF SS DIRECT EXPANSION FCU SCHEDULE (CLIN 0004)																					
	TAG	BUILDING LOCATED AT	AREA SERVED	TYPE	FAN			COOLING				ELECTRICAL				MAX NOISE (DBA)	SIZE HxWxD (MM)	WEIGHT (KG)	REMARKS			
					SA (LPS)	OA (LPS)	ESP (PA)	TOTAL (KW)	SENS. (KW)	DB (°C)	WB (°C)	DB (°C)	WB (°C)	POWER (W)	FLA (A)	MOP (A)	V/PH/Hz					
	FCU-H1	COF SS	H105 TELECOM	WALL MOUNTED	375	0	N/A	3.10	2.93	24.5	15.3	12.8	10.4	POWERED BY ACCU			200-208/1/60	55	293x798x230	10.0	1, 2, 3	
	FCU-H2	COF SS	H112 CLASSROOM	CEILING CONCEALED DUCTED	325	0	104	5.69	4.06	24.5	17.3	12.8	11.9	POWERED BY ACCU			200-208/1/60	60	298x1000x750	34.0	1, 2, 3	
	FCU-H3	COF SS	H113 OFFICE	WALL MOUNTED	80	0	N/A	1.30	1.15	24.5	16.2	12.8	11.0	POWERED BY ACCU			200-208/1/60	55	293x798x230	10.0	1, 2, 3	
	FCU-H4	COF SS	H122 OPEN WORKSTA./CONFERENCE	CEILING CONCEALED DUCTED	225	0	89.2	3.74	2.81	24.5	17.1	12.8	12.0	POWERED BY ACCU			200-208/1/60	60	298x1000x750	34.0	1, 2, 3	
E	FCU-H5	COF SS	H123/124/125/133/134/135 CO/1ST SGT/XO OFFICES	CEILING CONCEALED DUCTED	250	0	118.4	2.34	2.02	24.5	15.9	12.8	10.7	POWERED BY ACCU			200-208/1/60	60	298x1000x750	34.0	1, 2, 3	
	FCU-H6	COF SS	H132 OPEN WORKSTA./CONFERENCE	CEILING CONCEALED DUCTED	290	0	98.6	4.70	3.69	24.5	16.8	12.8	11.8	POWERED BY ACCU			200-208/1/60	60	298x1000x750	34.0	1, 2, 3	
	FCU-H7	COF SS	H205 TELECOM	WALL MOUNTED	375	0	N/A	3.21	3.05	24.5	15.3	12.8	10.2	POWERED BY ACCU			200-208/1/60	55	293x798x230	10.0	1, 2, 3	
	FCU-H8	COF SS	H212 OPEN WORKSTA./CONFERENCE	CEILING CONCEALED DUCTED	105	0	82.8	2.47	1.70	24.5	17.0	12.8	10.1	POWERED BY ACCU			200-208/1/60	60	298x1000x750	34.0	1, 2, 3	
	FCU-H9	COF SS	H213/214/215 CO/1ST SGT/XO OFFICES	CEILING CONCEALED DUCTED	80	0	89.1	1.46	1.34	24.5	16.0	12.8	8.7	POWERED BY ACCU			200-208/1/60	60	298x1000x750	34.0	1, 2, 3	
	FCU-H10	COF SS	H222 OPEN WORKSTA./CONFERENCE	CEILING CONCEALED DUCTED	215	0	85.1	4.24	3.42	24.5	17.3	12.8	11.3	POWERED BY ACCU			200-208/1/60	60	298x1000x750	34.0	1, 2, 3	
	FCU-H11	COF SS	H223/224/225/233/234/235 CO/1ST SGT/XO OFFICES	CEILING CONCEALED DUCTED	250	0	118.4	2.92	2.68	24.5	16.0	12.8	8.7	POWERED BY ACCU			200-208/1/60	60	298x1000x750	34.0	1, 2, 3	
D	FCU-H12	COF SS	H232 OPEN WORKSTA./CONFERENCE	CEILING CONCEALED DUCTED	265	0	93	5.04	4.07	24.5	16.9	12.8	11.0	POWERED BY ACCU			200-208/1/60	60	298x1000x750	34.0	1, 2, 3	
	NOTES: 1. FCU IS POWERED BY ACCU. 2. PROVIDE WITH MERV 8 FILTER. 3. PROVIDE WITH CONDENSATE PUMP.																					
C	SSAW DIRECT EXPANSION ACCU SCHEDULE (CLIN 0004)																					
	TAG	BUILDING LOCATED AT	AREA SERVED	TOTAL COOLING CAPACITY (KW)	AMBIENT AIR TEMP (C)	ELECTRICAL			SOUND (DBA)	DIMENSIONS HxWxD (MM)	WEIGHT (KG)	REMARKS										
						POWER (KW)	FLA (A)	MOCp (A)	V/PH/Hz													
	ACCU-D1	SSA WAREHOUSE	D107 SIPR ROOM	2.3	35	1.06	-	13.1	200-208/1/60	61	550x780x290	35.0 1										
	ACCU-D2	SSA WAREHOUSE	D109 TELECOM	4.8	35	1.52	-	13.0	200-208/1/60	61	550x780x290	40.0 1										
	NOTES: 1. PROVIDE WITH CORROSION PROTECTION, MINIMUM RATING 6000 HOURS OF ASTM B117-97 SALT SPRAY TEST.																					
B																						
A																						

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NOTES:

1. PROVIDE WITH CORROSION PROTECTION, MINIMUM RATING 6000 HOURS OF ASTM B117-97 SALT SPRAY TEST

NOTES:

1. PROVIDE WITH INTEGRAL BACKDRAFT DAMPER.
2. PROVIDE INTEGRAL DISCONNECT SWITCH.
3. PROVIDE WITH VFD.
4. FAN SHALL OPERATE ON A CONTINUOUS BASIS.
5. FAN SHALL OPERATE CONTINUOUSLY DURING BUSINESS HOURS.
6. FAN SHALL BE OF TYPE A SPARK RESISTANT CONSTRUCTION. FAN MOTOR SHALL NOT BE LOCATED IN AIR STREAM AND SHALL BE IN AN EXPLOSION PROOF ENCLOSURE.
7. INTERLOCK WITH FORKLIFT CHARGER(S).
8. INTERLOCK WITH SPACE OCCUPANCY SENSORS.
9. INTERLOCK WITH THERMOSTAT.

NOTES:
1. PROVIDE WITH INTEGRAL BACKDRAFT DAMPER.
2. PROVIDE INTEGRAL DISCONNECT SWITCH.
3. FAN SHALL OPERATE CONTINUOUSLY DURING BUSINESS HOURS

1. PROVIDE SIZE AS INDICATED ON DRAWINGS TO COINCIDE W/ DUCT.
2. PROVIDE INTEGRAL TRANSFORMER



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
P																					
O																					
N																					
M																					
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SSAW FLOW CONTROL VALVE SCHEDULE (CLIN 0004)										
VALVE NO.	EQUIPMENT	CAPACITY (LPM)	MAX PRESSURE DROP	VALVE SIZE (MM)	VALVE CV AT FULL FLOW	VALVE TYPE	CONFIGURATION	FAIL POSITION	MIN. CLOSEOFF PRESS. (KPA)	MATERIAL
			KPA							
CV-AHU-D1	AHU-D1	41.3	35	25	4.9	BALL	2-WAY	IN PLACE	690	NICKEL-PLATED BRASS BALL VALVE, TEFLON STEALS, ANGLE ROTATION: 0-90 DEG
CV-DOAS-D1	DOAS-D1	54.5	35	25	6.5	BALL	2-WAY	IN PLACE	690	

SSAW DUCT CLASSIFICATION SCHEDULE (CLIN 0004)												
SYSTEM	DUCT PRESSURE CLASS (PA)				SUPPLY				RETURN AIR/OUTSIDE AIR		DUCT PRESSURE TEST MM OF WATER COLUMN	NOTES
					ROUND/OVAL		RECTANGULAR		DUCT SEAL CLASS	DUCT LEAK CLASS		
	SUPPLY DUCT	RETURN DUCT	EXHAUST DUCT	OUTSIDE AIR DUCT	DUCT SEAL CLASS	DUCT LEAK CLASS	DUCT SEAL CLASS	DUCT LEAK CLASS				
AHU-D1	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
DOAS-D1	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-	-	500	-	-	-	-	A	24	50	[1]
EF-D1	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-D2	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-D3	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-D4	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-D5.1	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-D5.2	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-D5.3	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-D6	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-D7	-	-	-250	-	A	12	A	24	-	-	25	[1]

[1] TEST IN ACCORDANCE WITH UFGS 23 05 93, TESTING, ADJUSTING, AND BALANCING FOR HVAC AND THE PROCEDURE IN SMACNA HVAC AIR DUCT LEAKAGE TEST MANUAL

SSAW VIBRATION ISOLATOR SCHEDULE (CLIN 0004)				
EQUIPMENT	BASE TYPE	TYPE OF ISOLATOR	APPLICATION	DEFLECTION (MM.)
AHU-D1	A	3	PACKAGED AH, AC, H, AND V UNITS	38
DOAS-D1	A	3	PACKAGED AH, AC, H, AND V UNITS	38
ACCH-D1	A	4	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-D1	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-D2	A	1	REFRIGERANT MACHINES AND CHILLERS	19
EF-D1	A	3	EXHAUST FAN	38
EF-D2	A	3	EXHAUST FAN	38
EF-D3	A	3	EXHAUST FAN	38
EF-D4	A	3	EXHAUST FAN	38
EF-D5.1	A	3	EXHAUST FAN	38
EF-D5.2	A	3	EXHAUST FAN	38
EF-D5.3	A	3	EXHAUST FAN	38
EF-D6	A	3	EXHAUST FAN	38
EF-D7	A	3	EXHAUST FAN	38

BASE TYPE:
A. NO BASE, ISOLATORS ATTACHED DIRECTLY TO EQUIPMENT
B. STRUCTURAL STEEL RAILS OR BASE
C. CONCRETE INERTIA BASE
D. CURB-MOUNTED BASE

ISOLOATOR TYPE:
TYPE 1: PAD, RUBBER, OR GLASS FIBER
TYPE 2: RUBBER FLOOR ISOLATOR OR HANGER
TYPE 3: SPRING FLOOR ISOLATOR OR HANGER
TYPE 4: RESTRAINED SPRING ISOLATOR
TYPE 5: THRUST RESTANT

COF SS DUCT CLASSIFICATION SCHEDULE (CLIN 0004)												
SYSTEM	DUCT PRESSURE CLASS (PA)				SUPPLY				RETURN AIR/OUTSIDE AIR		DUCT PRESSURE TEST MM OF WATER COLUMN	NOTES
					ROUND/OVAL		RECTANGULAR		DUCT SEAL CLASS	DUCT LEAK CLASS		
	SUPPLY DUCT	RETURN DUCT	EXHAUST DUCT	OUTSIDE AIR DUCT	DUCT SEAL CLASS	DUCT LEAK CLASS	DUCT SEAL CLASS	DUCT LEAK CLASS				
DOAS-H1	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-	-	500	-	-	-	-	A	24	50	[1]
FCU-H1	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
FCU-H2	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
FCU-H3	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
FCU-H4	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
FCU-H5	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
FCU-H6	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
FCU-H7	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
FCU-H8	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
FCU-H9	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
FCU-H10	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
FCU-H11	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
FCU-H12	500	-	-	-	A	6	A	12	-	-	50	[1]
	-	-500	-	-	-	-	-	-	A	24	50	[1]
EF-H1	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-H2	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-H3	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-H4	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-H5	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-H6	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-H7	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-H8	-	-	-250	-	A	12	A	24	-	-	25	[1]
EF-H9	-	-	-250	-	A	12	A	24	-	-	25	[1]

[1] TEST IN ACCORDANCE WITH UFGS 23 05 93, TESTING, ADJUSTING, AND BALANCING FOR HVAC AND THE PROCEDURE IN SMACNA HVAC AIR DUCT LEAKAGE TEST MANUAL

COF SS VIBRATION ISOLATOR SCHEDULE (CLIN 0004)				
EQUIPMENT	BASE TYPE	TYPE OF ISOLATOR	APPLICATION	DEFLECTION (MM.)
DOAS-D1	A	3	PACKAGED AH, AC, H, AND V UNITS	38
FCU-H1	A	3	PACKAGED AH, AC, H, AND V UNITS	38
FCU-H2	A	3	PACKAGED AH, AC, H, AND V UNITS	38
FCU-H3	A	3	PACKAGED AH, AC, H, AND V UNITS	38
FCU-H4	A	3	PACKAGED AH, AC, H, AND V UNITS	38
FCU-H5	A	3	PACKAGED AH, AC, H, AND V UNITS	38
FCU-H6	A	3	PACKAGED AH, AC, H, AND V UNITS	38
FCU-H7	A	3	PACKAGED AH, AC, H, AND V UNITS	38
FCU-H8	A	3	PACKAGED AH, AC, H, AND V UNITS	38
FCU-H9	A	3	PACKAGED AH, AC, H, AND V UNITS	38
FCU-H10	A	3	PACKAGED AH, AC, H, AND V UNITS	38
FCU-H11	A	3	PACKAGED AH, AC, H, AND V UNITS	38
FCU-H12	A	3	PACKAGED AH, AC, H, AND V UNITS	38
ACCU-H1	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-H2	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-H3	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-H4	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-H5	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-H6	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-H7	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-H8	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-H9	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-H10	A	1	REFRIGERANT MACHINES AND CHILLERS	19
ACCU-H11	A	3	PACKAGED AH, AC, H, AND V UNITS	38
ACCU-H12	A	4	REFRIGERANT MACHINES AND CHILLERS	19
EF-H1	A	3	EXHAUST FAN	38
EF-H2	A	3	EXHAUST FAN	38
EF-H3	A	3	EXHAUST FAN	38
EF-H4	A	3	EXHAUST FAN	38
EF-H5	A	3	EXHAUST FAN	38
EF-H6	A	3	EXHAUST FAN	38
EF-H7	A	3	EXHAUST FAN	38
EF-H8	A	3	EXHAUST FAN	38
EF-H9	A	3	EXHAUST FAN	38

BASE TYPE:
A. NO BASE, ISOLATORS ATTACHED DIRECTLY TO EQUIPMENT
B. STRUCTURAL STEEL RAILS OR BASE
C. CONCRETE INERTIA BASE
D. CURB-MOUNTED BASE

ISOLOATOR TYPE:
TYPE 1: PAD, RUBBER, OR GLASS FIBER
TYPE 2: RUBBER FLOOR ISOLATOR OR HANGER
TYPE 3: SPRING FLOOR ISOLATOR OR HANGER
TYPE 4: RESTRAINED SPRING ISOLATOR
TYPE 5: THRUST RESTANT



US Army Corps
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DATE

MARK

DESCRIPTION

DESIGNED BY:
L TSANG

DRAWN BY:
JAPAN ENGINEER DISTRICT
OKINAWA AREA OFFICE
APO AP 96343-5132

ISSUED BY:
O CRABTREE

SUBMITTED BY:
O CRABTREE

SCALE:
AS SHOWN

SIZE:
ANSI D

US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA AREA OFFICE
APO AP 96343-5132

DESIGN PARTNERS INCORPORATED
1580 MAKALOA STREET, SUITE 1100
HONOLULU, HAWAII 96814

APO AP 96343-5010
FY20 MCA PN77833 VEHICLE MAINTENANCE SHOP

MECHANICAL SCHEDULES

SHEET ID

M-608

INTERMEDIATE DESIGN

SSAW - UNCONDITIONED AREAS / WAREHOUSE - ROOM AIR BALANCE TABLE (CLIN 0004)																
VAV TAG	ROOM	NAME	AREA (M^2)	DESIGN SUPPLY AIR (L/s)	MIN SUPPLY AIR (L/s)	MIN RETURN AIR (L/s)	MIN OUTSIDE AIR (L/s)	DESIGN OA (L/s)	UNCONDITIONED OA (L/s)	MIN EXHAUST AIR (L/s)	DESIGN EXHAUST AIR (L/s)	TRANSFER IN (L/s)	TRANSFER OUT (L/s)	BALANCE (L/s)	SERVED BY	REMARKS
N/A	D116	FORKLIFT CHARGE STATION	17.41	0	0	0	0.0	0	180.0	180	180	0	0	0	EF-D4	
N/A	D118	WAREHOUSE & RECEIVING/ISSUE BAY	1197.13	0	0	0	0.0	0	9000	9000	9000	0	0	0	EF-D5.1/D5.2/D5.3	
N/A	D120	CUSTOMER ISSUE BAY	167.22	0	0	0	0.0	0	1260.0	1260	1260	0	0	0	EF-D6	
N/A	D121	RECEIVING/ISSUE BAY	SEE D118 (REDUNDANT AREA)											0		
N/A	D122	TURN-IN BAY	SEE D118 (REDUNDANT AREA)											0		
N/A	D127	MECHANICAL YARD	44	0	0	0	0.0	0	0.0	0	0	0	0	0	N/A	
		SUMMED AIR FLOWS		0	0	0	0		10440	10440	10440	0	0	0		
BUILDING PRESSURIZATION (%)															0%	



US Army Corps
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US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96376-5132	DESIGNED BY:	ISSUE DATE:
	I. TSANG I. TSANG I. TSANG	SOLICITATION NO.: CONTRACT NO.: SCALE:
DESIGN PARTNERS INCORPORATED 1580 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814	CHECKED BY:	W912HW-17-D-0003 AS SHOWN
	O. CRABTREE SUBMITTED BY: SIZE:	CRABTREE AS SHOWN
	ANSI D	

SHEET ID
M-609

P —
 O —
 N —
 M —
 L —
 K —
 J —
 I —
 H —
 G —
 F —
 E —
 D —
 C —
 B —
 A

0	25	DOAS-H1	
60	80	FCU-H8/DOAS-H1	
15	10	FCU-H5/DOAS-H1	
15	10	FCU-H5/DOAS-H1	
15	10	FCU-H5/DOAS-H1	
0	-10	EF-H4	
0	-10	EF-H4	
410	145		20.3%
0	25	DOAS-H1	
0	-35	EF-H5	
0	-35	EF-H5	
0	-35	EF-H5	
0	25	FCU-H7/DOAS-H1	
0	25	DOAS-H1	
225	20	FCU-H8/DOAS-H1	
15	10	FCU-H9/DOAS-H1	
15	10	FCU-H9/DOAS-H1	
15	10	FCU-H9/DOAS-H1	
0	-40	EF-H6	
0	-40	EF-H6	
0	25	DOAS-H1	
60	80	FCU-H10/DOAS-H1	
15	10	FCU-H11/DOAS-H1	
15	10	FCU-H11/DOAS-H1	
15	10	FCU-H11/DOAS-H1	
0	-10	EF-H7	
0	-10	EF-H7	
0	25	DOAS-H1	
60	80	FCU-H12/DOAS-H1	
15	10	FCU-H11/DOAS-H1	
15	10	FCU-H11/DOAS-H1	
15	10	FCU-H11/DOAS-H1	
0	-10	EF-H8	
0	-10	EF-H8	
480	170		23.0%

0 - 75

76 - 125

126 - 175

176 - 250

CEILING EXHAUST REGISTER: PERFECT

LPS RANGE - RETURN AND EXHAUST

0 - 50

51 - 100

101 - 150

151 - 200

201 - 300

301 - 400

401 - 500

DIFFUSER AND REGISTER KEY

SUPPLY:

THROW DIRECTIONS

DIFFUSER TYPE → S4-200 ←

OUTSIDE:

THROW DIRECTIONS

DIFFUSER TYPE → O4-200 ←

INTERMEDIATE DESIGN

1. THE DOAS UNIT SHALL BE TURNED ON AND OFF VIA THE DDC SYSTEM WHEN THE H-O-A SWITCH IS PLACED IN THE "AUTO" MODE. THE UNIT SHALL BE OPERATED MANUALLY WHEN THE H-O-A SWITCH IS PLACED IN THE "HAND" MODE.
2. UPON ACTIVATION OF THE DUCT SMOKE DETECTOR, A SIGNAL SHALL BE RELAYED TO THE FIRE ALARM SYSTEM AND SHALL SHUTDOWN THE DOAS UNIT.
3. THE DOAS SHALL PROVIDE ROOM DIRECT OUTSIDE AIR. HOT GAS REHEAT VALVE SHALL MODULATE TO MAINTAIN REHEAT COIL SUPPLY AIR TEMPERATURE OF 75°F.
4. ACTIVATION OF THE ATFP EMERGENCY HVAC SHUTDOWN SWITCH SHALL AUTOMATICALLY SHUT DOWN THE DOAS AND SHALL AUTOMATICALLY CLOSE THE LOW LEAKAGE DAMPERS ON THE OA INTAKE SIDE.



INTERMEDIATE DESIGN

1.3 SYSTEM SCHEDULER OUTPUT DETERMINATION

FOR CONTROLLING AN AIR HANDLER, INTERPRET A SNVT INPUT OF OC_STANDBY AS WARM-UP-COOL-DOWN IF THE SEQUENCE OF OPERATION SUPPORTS THAT MODE, OTHERWISE INTERPRET OC_STANDBY AS OCCUPIED. FOR TERMINAL UNITS, INTERPRET OC_STANDBY AS OCCUPIED.

1.3.1 AIR HANDLER OCCUPANCY OUTPUT

2.1.2 OFF-COOL SWITCH

2.1.2.1 COOL

2.1.2.2 OFF

2.1.3 OCCUPANCY MODES

2.1.3.1 OCCUPIED

2.1.3.2 UNOCCUPIED

2.1.4 SAFETIES

RUN THE UNIT SUBJECT TO THE UNIT MANUFACTURER'S SAFETIES.

2.1.5 ZONE TEMPERATURE CONTROL

b. IN THE UNOCCUPIED MODE THE ZONE TEMPERATURE SETPOINT (ZN-T-SP-UNOCC) MUST BE AT THE CONFIGURED SETPOINT (ZN-T-SP-UNOCC) AS INDICATED.

c. CYCLE THE FAN, COOLING UNIT, HEATING UNIT WITH THE DDO HARDWARE, IN ACCORDANCE WITH THE HEAT-COOL SWITCH SETTING, TO MAINTAIN ZONE TEMPERATURE (ZN-T) AT SETPOINT (ZN-T-SP).

1.4 AIR HANDLER SYSTEM SCHEDULING

a. BIND THE AHU OCCUPANCY OUTPUT SNVT FROM THE SYSTEM SCHEDULER TO THE DDC HARDWARE THAT EXECUTES THE OCCUPANCY MODE DETERMINATION PART OF THE AIR HANDLER SEQUENCE OF OPERATION

b. BIND THE TERMINAL UNIT OCCUPANCY OUTPUT SNVT FROM THE SYSTEM SCHEDULER TO EACH AHU-DEPENDENT TERMINAL UNIT.

1.5 STAND-ALONE TERMINAL UNIT SCHEDULING

BIND THE TERMINAL UNIT OCCUPANCY OUTPUT FROM THE SYSTEM
 SCHEDULER TO THE DDC HARDWARE THAT EXECUTES THE OCCUPANCY
 MODE DETERMINATION PART OF THE TERMINAL UNIT SEQUENCE OF
 OPERATION.

2. SEQUENCES OF OPERATION FOR AIR HANDLING UNITS

2.1 ALL-AIR SMALL PACKAGE UNITARY SYSTEM

INSTALL DDC HARDWARE TO PERFORM THIS SEQUENCE OF OPERATION AND TO PROVIDE SNVT INPUTS AND OUTPUTS AS SPECIFIED AND SHOWN ON THE POINTS SCHEDULE.

UNLESS OTHERWISE SPECIFIED, ALL MODULATING CONTROL MUST BE PROPORTIONAL-INTEGRAL (PI) CONTROL.

2.1.1 FAN ON-AUTO SWITCH

2.1.1.1 ON

WITH THE THERMOSTAT FAN ON-AUTO SWITCH IN THE ON POSITION, THE DDC HARDWARE MUST START AND CONTINUOUSLY RUN THE FAN.

2.1.1.2 AUTO

WITH THE THERMOSTAT FAN ON-AUTO SWITCH IN THE AUTO POSITION, THE DDC HARDWARE OPERATES THE FAN ACCORDING TO HEAT-OFF-COOL SWITCH.

1.1 SYSTEM MODE

1.2 SYSTEM SCHEDULER REQUIREMENTS

THE SYSTEM SCHEDULER FUNCTIONALITY MUST RESIDE IN EITHER A PIECE OF DDC HARDWARE DEDICATED TO THIS FUNCTIONALITY OR IN THE DDC HARDWARE CONTROLLING THE SYSTEM AHU. A SINGLE PIECE OF DDC HARDWARE DEDICATED TO SCHEDULING (PERFORMING NO OTHER CONTROL FUNCTIONALITY) MAY CONTAIN MULTIPLE SYSTEM SCHEDULERS. PROVIDE A UNIQUE SYSTEM SCHEDULER FOR: EACH AHU INCLUDING ITS ASSOCIATED TERMINAL UNITS, AND EACH STAND-ALONE TERMINAL UNIT (THOSE NOT DEPENDENT UPON AHU SERVICE) OR GROUP OF STAND-ALONE TERMINAL UNITS ACTING ACCORDING TO A COMMON SCHEDULE. EACH SYSTEM SCHEDULER MUST PROVIDE THE FOLLOWING FUNCTIONALITY:

1.2.1 SCHEDULED OCCUPANCY INPUT

ACCEPT NETWORK VARIABLE OF TYPE SNVT_OCCUPANCY. SUPPORT THE FOLLOWING POSSIBLE VALUES: OC_STANDBY, OC_OCCUPIED AND OC_UNOCCUPIED.

1.2.2 OCCUPANCY OVERRIDE INPUT

ACCEPT NETWORK VARIABLE OF TYPE SNVT_OCCUPANCY. SUPPORT THE FOLLOWING POSSIBLE VALUES: OC_STANDBY, OC_OCCUPIED, OC_UNOCCUPIED, AND OC_NUL.

1.2.3 AIR HANDLER OCCUPANCY OUTPUT

FOR A SYSTEM SCHEDULER FOR A SYSTEM CONTAINING AN AIR HANDLER, OUTPUT ONE OR MORE SNVTS INDICATING THE DESIRED OCCUPANCY STATUS AS ONE OF THE FOLLOWING POSSIBLE VALUES: WARM-UP-COOL-DOWN (WHEN REQUIRED BY THE AHU SEQUENCE OF OPERATION), OCCUPIED AND UNOCCUPIED.

1.2.4 TERMINAL UNIT OCCUPANCY OUTPUT

FOR A SYSTEM SCHEDULER FOR A STAND-ALONE TERMINAL UNIT, A GROUP OF STAND-ALONE TERMINAL UNITS ACTING ACCORDING TO A COMMON SCHEDULE, OR A GROUP OF TERMINAL UNITS SERVED BY A SINGLE AIR HANDLER, OUTPUT ONE OR MORE SNVTS INDICATING THE DESIRED OCCUPANCY STATUS AS ONE OF THE FOLLOWING POSSIBLE VALUES: OCCUPIED AND UNOCCUPIED.

1.2.5 DEFAULT SCHEDULE

INCORPORATE A 24-HOUR 7-DAY DEFAULT SCHEDULE AS SHOWN ON THE DRAWINGS WHICH MAY BE ACTIVATED AND DEACTIVATED BY THE SYSTEM SCHEDULER LOGIC.

1.2.6 COMMUNICATION DETERMINATION

DETERMINE THE TIME ELAPSED BETWEEN RECEIPTS OF THE SCHEDULED OCCUPANCY INPUT SNVT, AND USE THIS ELAPSED TIME TO ACTIVATE AND DEACTIVATE THE DEFAULT SCHEDULE AS SPECIFIED. (THIS PROVIDES THE CAPABILITY FOR THE SYSTEM SCHEDULER TO USE ITS DEFAULT SCHEDULE IF IT LOSES COMMUNICATION WITH THE UMCS).



12.3.4.3DDC HARDWARE RESET

DDC HARDWARE RESET OF ALL PROOFS AND SAFETIES SHALL BE VIA A LOCAL BINARY PUSH-BUTTON (RST-BUT) INPUT TO THE DDC HARDWARE, VIA A REMOTE COMMAND TO THE DDC HARDWARE VIA SNVT OR BOTH (WHERE THE CONTRACTOR PROVIDES BOTH RESET FUNCTIONS AND THE OPERATOR CAN USE EITHER ONE TO PERFORM THE RESET), AS SHOWN ON THE POINTS SCHEDULE DRAWING.

2.4.4SYSTEM ENABLE AND LOOP ENABLE

2.4.4.1OCCUPIED MODE

THE SUPPLY FAN SHALL BE ENABLED (SYS-ENA) AND COMMANDED TO RUN (SF-SS). ALL CONTROL LOOPS SHALL BE ENABLED.

2.4.4.2UNOCCUPIED MODE

WHILE THE BUILDING TEMPERATURE (BLDG-T) IS ABOVE THE LOW LIMIT SETPOINT (BLDG-T-LL) ALL CONTROL LOOPS SHALL BE DISABLED AND THE SUPPLY FAN SHALL NOT RUN. WHEN BLDG-T DROPS BELOW BLDG-T-LL (WITH A 5 DEGREES F DEADBAND) THE SUPPLY FAN SHALL BE ENABLED (SYS-ENA) AND COMMANDED TO RUN (SF-SS), THE SUPPLY DUCT STATIC PRESSURE CONTROL LOOPS SHALL BE ENABLED. THE MINIMUM OUTSIDE AIR FLOW CONTROL, MIXED AIR TEMPERATURE CONTROL, AND COOLING COIL CONTROL LOOPS SHALL BE DISABLED.

2.4.5FAN CAPACITY CONTROL

2.4.5.1SUPPLY DUCT STATIC PRESSURE CONTROL

WHEN THIS LOOP IS ENABLED THE DDC HARDWARE SHALL MODULATE THE SUPPLY FAN VARIABLE FREQUENCY DRIVE UNIT TO MAINTAIN THE DUCT STATIC PRESSURE (SA-P) AT SETPOINT (SA-P-SP) AS SHOWN, AS MEASURED BY THE DUCT STATIC PRESSURE TAP AND SENSOR AS SHOWN. WHEN THIS LOOP IS DISABLED, THE DDC HARDWARE CAPACITY MODULATION OUTPUT TO THE VFD SHALL BE ZERO PERCENT.

2.4.6MINIMUM OUTSIDE AIR FLOW CONTROL

WHEN THIS LOOP IS ENABLED THE DDC HARDWARE SHALL MODULATE THE MINIMUM OUTSIDE AIR DAMPER TO MAINTAIN THE MINIMUM OA VOLUMETRIC FLOW (MINOA-F) AT SETPOINT (MINOA-F-SP) AS SHOWN. WHEN THIS LOOP IS DISABLED, THE MINIMUM OUTSIDE AIR DAMPER SHALL BE CLOSED.

2.4.7COOLING COIL CONTROL

WHEN THIS LOOP IS ENABLED THE DDC HARDWARE SHALL MODULATE THE COOLING COIL VALVE TO MAINTAIN THE SUPPLY AIR TEMPERATURE (SA-T) SETPOINT (SA-T-SP) AS SHOWN. WHEN THIS LOOP IS DISABLED, THE COOLING COIL VALVE SHALL BE CLOSED.

2.4.8TEMPERATURE CONTROL

THE CHILLED WATER CONTROL VALVE SHALL MODULATE TO MAINTAIN THE SUPPLY AIR TEMPERATURE AT 12.7 °C. A TEMPERATURE RESET SCHEDULE SHALL BE PROGRAMMED INTO THE SYSTEM AS FOLLOWS:

ON A FALL IN RETURN AIR TEMPERATURE BELOW 24 °C, THE SUPPLY AIR TEMPERATURE MAINTAINED BY THE CHW CONTROL VALVE SHALL BE RESET BASED ON THE FOLLOWING RETURN AIR TEMPERATURES AND HUMIDITY SET POINT SCHEDULES:

HUMIDITY	SUPPLY AIR SETPOINT
50%	6.7 DEGREE C
45%≤RH<50%	13.9 DEGREE C
40%≤RH<45%	15 DEGREE C
RH≤45%	16 DEGREE C
RETURN AIR TEMPERATURE	SUPPLY AIR SETPOINT
24.4 DEGREE C	6.7 DEGREE C
24.4 DEGREE C ≤T<24.4 DEGREE C	13.9 DEGREE C
22.2 DEGREE C ≤T<24.4 DEGREE C	15 DEGREE C
21.1 DEGREE C ≤T<24.4 DEGREE C	16.1 DEGREE C
NOTE: DEHUMUDIFICATION MODE SHALL OVERRIDE SUPPLY AIR TEMPERATURE RESET SHOULD THE HUMIDITY INCREASE AND RETURN AIR DECREASE.	

2.2.6CONTROLS

THE DOAS AHU SHALL BE TURNED ON AND OFF VIA THE DDC SYSTEM WHEN THE H-O-A SWITCH IS PLACED IN THE "AUTO" MODE. THE UNIT SHALL BE OPERATED MANUALLY WHEN THE H-O-A SWITCH IS PLACED IN THE "HAND" MODE. ALL AIR HANDLERS AND THEIR CORRESPONDING TERMINAL UNITS SHALL OPERATE ON THE TIME SCHEDULE PROGRAMMED INTO THE DDC SYSTEM.

AT START-UP, THE DDC SYSTEM SHALL OPEN THE CHILLED WATER FLOW CONTROL VALVE TO ALLOW FLOW THROUGH THE DOAS COOLING COIL. THE VFD SHALL BE PROGRAMMED TO SOFT START THE DOAS MOTOR AT START-UP TO PREVENT OVER-PRESSURIZATION OF THE DUCTWORK.

THE CHILLED WATER CONTROL VALVE SHALL MODULATE TO MAINTAIN THE SUPPLY AIR TEMPERATURE AT 12.7 °C DIRECTLY AFTER THE COOLING COIL. THE OUTSIDE AIR SHALL BE TEMPERED TO PROVIDE NEUTRAL VENTILATION AIR TO THE SPACE BY THE WRAP AROUND HEAT PIPE. DOAS SHALL OPERATE INDEPENDENTLY OF AHU'S.

THE VFD IN THE DOAS'S SHALL BE USED ONLY FOR SOFT START. THE VFD WILL BE SET TO SUPPLY CONSTANT AIRFLOW THROUGH DOAS. UPON ACTIVATION OF THE DUCT SMOKE DETECTOR, A SIGNAL SHALL BE RELAYED TO THE FIRE ALARM SYSTEM AND THE FIRE ALARM SYSTEM SHALL SHUTDOWN THE AHU. COORDINATE WITH FIRE ALARM CONTRACTOR.

2.3SINGLE ZONE DX COOLING COILS

INSTALL DDC HARDWARE TO PERFORM THIS SEQUENCE OF OPERATION AND TO PROVIDE SNVT INPUTS AND OUTPUTS AS SPECIFIED AND SHOWN ON THE POINTS SCHEDULE. UNLESS OTHERWISE SPECIFIED, ALL MODULATING CONTROL MUST BE PROPORTIONAL-INTEGRAL (PI) CONTROL.

2.3.1HAND-OFF-AUTO SWITCH

SUPPLY FAN MOTOR STARTER MUST ACCEPT A FIRE ALARM PANEL (FAP) SIGNAL THAT TAKES PRECEDENCE OVER ALL OTHER STARTER INPUTS AND SWITCHES AND START THE FAN. THE FAN MOTOR STARTER MUST ACCEPT AN OCCUPANT ACCESSIBLE EMERGENCY SHUTOFF SWITCH AS INDICATED. THE SUPPLY FAN MOTOR STARTER MUST HAVE AN H-O-A SWITCH:

2.3.1.1HAND

WITH THE H-O-A SWITCH IN HAND POSITION, THE SUPPLY FAN STARTS AND RUNS CONTINUOUSLY, SUBJECT TO SAFETIES.

2.3.1.2OFF

WITH THE H-O-A SWITCH IN OFF POSITION, THE SUPPLY FAN STOPS.

2.3.1.3AUTO

WITH THE H-O-A SWITCH IN AUTO POSITION, THE SUPPLY FAN RUNS SUBJECT TO THE SUPPLY FAN START/STOP (SF-SS) COMMAND AND SAFETIES.

2.2OCCUPANCY MODES

OBTAIN THE SYSTEM'S OCCUPANCY MODE INPUT FROM THE SYSTEM SCHEDULER AS SPECIFIED AND INDICATED. OPERATE THE SYSTEM IN ONE OF THE FOLLOWING MODES: OCCUPIED, UNOCCUPIED.

2.3.3SYSTEM ENABLE AND LOOP ENABLE

2.3.3.1OCCUPIED MODE

ENABLE THE SUPPLY FAN (SYS-ENA) AND COMMAND TO RUN (SF-SS) AND ENABLE ALL CONTROL LOOPS.

2.3.2UNOCCUPIED MODE

WHILE THE BUILDING TEMPERATURE (BLDG-T) IS ABOVE THE LOW LIMIT SETPOINT (BLDG-T-LL) DISABLE ALL CONTROL LOOPS AND THE SUPPLY FAN DOES NOT RUN. WHEN BLDG-T DROPS BELOW BLDG-T-LL (WITH A 5 DEGREES F DEADBAND) ENABLE THE SUPPLY FAN (SYS-ENA) AND COMMAND TO RUN (SF-SS) AND ENABLE THE HEATING COIL TEMPERATURE CONTROL LOOP. DISABLE THE OUTSIDE AIR FLOW CONTROL, ECONOMIZER DAMPER CONTROL, AND DX COOLING COIL CONTROL LOOPS.

2.3.4PROOFS AND SAFETIES

THE SUPPLY FAN AND ALL DDC HARDWARE CONTROL LOOPS ARE SUBJECT TO PROOFS AND SAFETIES. SAFETIES MUST BE DIRECT-HARDWIRE INTERLOCKED TO THE FAN STARTER CIRCUIT AS INDICATED. DDC HARDWARE MUST MONITOR ALL PROOFS AND SAFETIES AND FAILURE OF ANY PROOF OR ACTIVATION OF ANY SAFETY RESULT IN ALL CONTROL LOOPS BEING DISABLED AND THE AHU FAN BEING COMMANDED OFF UNTIL RESET.

2.3.4.1PROOFS

SUPPLY FAN STATUS (PROOF) (SF-S)

2.3.4.2SAFETIES

a. SUPPLY AIR SMOKE (SA-SMK)

2.3.4.3DDC HARDWARE

DDC HARDWARE RESET OF ALL PROOFS AND SAFETIES SHALL BE VIA A LOCAL BINARY PUSH-BUTTON (RST-BUT) INPUT TO THE DDC HARDWARE, VIA A REMOTE COMMAND TO THE DDC HARDWARE VIA SNVT OR BOTH (WHERE THE CONTRACTOR PROVIDES BOTH RESET FUNCTIONS AND THE OPERATOR CAN USE EITHER ONE TO PERFORM THE RESET), AS SHOWN ON THE POINTS SCHEDULE DRAWING.

2.3.5MINIMUM OUTSIDE AIR FLOW CONTROL

WHEN THIS LOOP IS ENABLED THE DDC HARDWARE SHALL OPEN THE 2-POSITION MINIMUM OUTSIDE AIR DAMPER TO INTRODUCE THE MINIMUM OUTSIDE AIR FLOW QUANTITY AS SHOWN. WHEN THIS LOOP IS DISABLED, THE MINIMUM OUTSIDE AIR DAMPER SHALL BE CLOSED.

2.4VARIABLE AIR VOLUME SYSTEM

INSTALL DDC HARDWARE TO PERFORM THIS SEQUENCE OF OPERATION AND TO PROVIDE SNVT INPUTS AND OUTPUTS AS SPECIFIED AND SHOWN ON THE POINTS SCHEDULE. UNLESS OTHERWISE SPECIFIED, ALL MODULATING CONTROL SHALL BE PROPORTIONAL-INTEGRAL (PI) CONTROL.

2.4.1HAND-OFF-AUTO SWITCHES

SUPPLY FAN VARIABLE FREQUENCY DRIVE (VFD) UNIT SHALL ACCEPT A FIRE ALARM PANEL (FAP) SIGNAL THAT TAKES PRECEDENCE OVER ALL OTHER VFD INPUTS AND SWITCHES AND SHALL CAUSE THE VFD TO RUN AT 100 PERCENT SPEED. THE VFD SHALL ACCEPT AN OCCUPANT ACCESSIBLE EMERGENCY SHUTOFF SWITCH AS SHOWN. THE SUPPLY FAN VARIABLE FREQUENCY DRIVE (VFD) UNIT SHALL HAVE AN INTEGRAL H-O-A SWITCH:

2.4.1.1HAND

WITH THE H-O-A SWITCH IN HAND POSITION, THE SUPPLY FAN SHALL START AND RUN CONTINUOUSLY, SUBJECT TO SAFETIES. FAN SPEED SHALL BE UNDER MANUAL-OPERATOR CONTROL.

2.4.1.2OFF

WITH THE H-O-A SWITCH IN OFF POSITION, THE SUPPLY FAN SHALL STOP.

2.4.1.3AUTO

WITH THE H-O-A SWITCH IN AUTO POSITION, THE SUPPLY FAN SHALL RUN SUBJECT TO THE SUPPLY FAN START/STOP SIGNAL (SF-SS) AND SAFETIES. FAN SPEED SHALL BE UNDER CONTROL OF THE DDC HARDWARE.

2.4.2OCCUPANCY MODES

THE SYSTEM SHALL OBTAIN ITS OCCUPANCY MODE INPUT FROM THE SYSTEM SCHEDULER AS SPECIFIED AND SHOWN. THE SYSTEM SHALL OPERATE IN ONE OF THE FOLLOWING MODES: OCCUPIED, UNOCCUPIED.

2.4.3PROOFS AND SAFETIES

THE SUPPLY FAN, AND ALL DDC HARDWARE CONTROL LOOPS SHALL BE SUBJECT TO PROOFS AND SAFETIES. SAFETIES SHALL BE DIRECT-HARDWIRE INTERLOCKED TO THE VFD AS SHOWN. DDC HARDWARE SHALL MONITOR ALL PROOFS AND SAFETIES AND FAILURE OF ANY PROOF OR ACTIVATION OF ANY SAFETY SHALL RESULT IN ALL CONTROL LOOPS BEING DISABLED AND THE AHU FAN BEING COMMANDED OFF UNTIL RESET.

2.4.3.1PROOFS

a. SUPPLY FAN STATUS (SF-S)

2.4.2SAFETIES

a. SUPPLY AIR DUCT PRESSURE HIGH LIMIT (SA-P-HL)

b. SUPPLY AIR SMOKE (SA-SMK)

3SEQUENCES OF OPERATION FOR TERMINAL UNITS

3.1ZONE TEMPERATURE CONTROL - COOLING-ONLY VAV BOX

INSTALL DDC HARDWARE TO PERFORM THIS SEQUENCE OF OPERATION AND TO PROVIDE SNVT INPUTS AND OUTPUTS AS SPECIFIED AND SHOWN ON THE POINTS SCHEDULE. UNLESS OTHERWISE SPECIFIED, ALL MODULATING CONTROL SHALL BE PROPORTIONAL-INTEGRAL (PI) CONTROL.

3.1.1OCCUPANCY MODES

3.1.1.1OCCUPIED

THE VAV BOX DDC HARDWARE SHALL BE IN THE OCCUPIED MODE WHEN THE INPUT FROM THE SYSTEM SCHEDULER (SYS-OCC) IS OCCUPIED.

3.1.1.2UNOCCUPIED

THE VAV BOX DDC HARDWARE MUST BE IN THE UNOCCUPIED MODE INPUT FROM THE SYSTEM SCHEDULER (SYS-OCC) IS UNOCCUPIED.

3.1.1.3SAFETIES

THIS SYSTEM HAS NO SAFETIES.

3.1.2ZONE TEMPERATURE CONTROL

3.1.2.1OCCUPIED MODE

IN THE OCCUPIED MODE THE ZONE TEMPERATURE SETPOINT (ZN-T-SP) MUST BE AT THE CONFIGURED SETPOINT OR AT THE OCCUPANT-ADJUSTABLE SETPOINT VIA THE WALL-MOUNTED THERMOSTAT, AS SHOWN. THE DDC HARDWARE MUST MODULATE THE VAV BOX DAMPER TO MAINTAIN VAV BOX SUPPLY AIR FLOW (VAV-SA-F) AT SETPOINT AS MEASURED BY A MULTI-POINT FLOW SENSING ELEMENT AT THE INLET TO THE VAV BOX. SEQUENCEAS INDICATED: UPON A RISE IN ZONE TEMPERATURE (ZN-T) ABOVE ZONE SETPOINT (ZN-T-SP), SUBJECT TO THE ZONE TEMPERATURE SETPOINT DEADBAND AS INDICATED, ADJUST THE AIRFLOW SETPOINT BETWEEN MINIMUM AND MAXIMUM FLOW BASED ON THE DIFFERENCE BETWEEN ZONE TEMPERATURE AND ZONE TEMPERATURE SETPOINT AS INDICATED.

3.1.2.2UNOCCUPIED MODE

IN THE UNOCCUPIED MODE THE VAV BOX DAMPER SHALL BE AT ITS MINIMUM POSITION.

3.1.3ZONE TEMPERATURE CONTROL

a. IN THE OCCUPIED MODE THE ZONE TEMPERATURE SETPOINT (ZN-T-SP) SHALL BE AT THE CONFIGURED SETPOINT OR AT THE OCCUPANT-ADJUSTABLE SETPOINT VIA THE WALL-MOUNTED THERMOSTAT, AS SHOWN.

b. IN THE UNOCCUPIED MODE THE ZONE TEMPERATURE SETPOINT (ZN-T-SP) SHALL BE AT THE CONFIGURED SETPOINT AS SHOWN.

c. THE DDC HARDWARE SHALL MODULATE THE VAV BOX DAMPER TO MAINTAIN VAV BOX SUPPLY AIR FLOW (VAV-SA-F) AT SETPOINT AS MEASURED BY A MULTI-POINT FLOW SENSING ELEMENT AT THE INLET TO THE VAV BOX. SEQUENCING SHALL BE AS SHOWN: UPON A RISE IN ZONE TEMPERATURE ABOVE ZONE TEMPERATURE SETPOINT (ZN-T-SP), SUBJECT TO THE ZONE TEMPERATURE SETPOINT DEADBAND AS SHOWN, THE AIRFLOW SETPOINT SHALL BE ADJUSTED BETWEEN MINIMUM AND MAXIMUM FLOW BASED ON THE DIFFERENCE BETWEEN ZONE TEMPERATURE AND ZONE TEMPERATURE SETPOINT AS SHOWN. UPON A FALL IN ZONE TEMPERATURE BELOW ZONE TEMPERATURE SETPOINT, SUBJECT TO THE DEADBAND AS SHOWN, THE AIRFLOW SHALL BE MAINTAINED AT A FIXED AIR FLOW SETPOINT (WITH A SETTING INDEPENDENT OF THE COOLING MINIMUM AIR FLOW), AND THE HEATING VALVE SHALL MODULATE TOWARDS OPEN OR THE STAGED ELECTRIC RESISTANCE HEATING COIL(S) SHALL CYCLE ON IN SEQUENCE.

US Army Corps of Engineers®

ISSUE DATE:

DESIGNED BY:

DRAWN BY:

CHECKED BY:

SUBMITTED BY:

SIZE:

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APO AP 96343-5010

FY20 MCA PN77833 VEHICLE MAINTENANCE SHOP

MECHANICAL SEQUENCE OF OPERATIONS

SHEET ID

MI-706

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MECHANICAL SEQUENCE OF OPERATIONS

SCALE: NTS

INTERMEDIATE DESIGN

3.6 EXHAUST FAN SEQUENCE OF OPERATIONS

EXHAUST FAN SHALL BE VARIABLE VOLUME AND

EXHAUST FAN SHALL BE VARIABLE VOLUME AND SHALL BE ENABLED AS SPECIFIED BY THE SYSTEM SCHEDULER, 24-HR OPERATION FOR BOTH DOAS AND EF. EF SHALL VARY IN WITH DOAS. DOAS UNIT SHALL PROVIDE ADEQUATE OUTSIDE AIR FOR EXHAUST FAN MAKE-UP (SEE AIR BALANCE SCHEDULES).

3.7 EMERGENCY SHUTDOWN SEQUENCE OF OPERATIONS

UPON A SIGNAL TO START THE CHILLER, THE UNIT'S CONTROL SYSTEM WILL FIRST CHECK THE INCOMING POWER TO ENSURE THAT THERE IS NO PROBLEMS WITH SINGLE PHASING, PHASE REVERSAL, OR UNBALANCED VOLTAGES.

AFTER A 60 SECOND TIME DELAY AND IF EVERY SAFETY AND INTERLOCK IS CONFIRMED, THE CHILLER COMPRESSOR MOTOR IS THEN STARTED. THE STARTUP CYCLE FROM SIGNAL TO START TO ACTUAL COMPRESSOR MOTOR START SHALL BE ACCOMPLISHED WITHIN A MAXIMUM OF TWO MINUTES.

ONCE THE CHILLER IS STARTED, THE STARTER WILL INITIATE ITS STARTING SEQUENCE AND WILL BE CONFIRMED TO BE IN OPERATION. MONITORING CIRCUITS WILL THEN MONITOR CURRENT LIMITS, CONDENSER LIMITS, EVAPORATOR LIMITS, AND SURGE CONDITION FOR PROPER OPERATION.

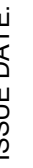
THE CONTROL SYSTEM WILL STOP THE CHILLER FOR THE FOLLOWING REASONS: LOW LOAD, SET POINT DEVIATION, LATCHING SAFETY OR MANUAL SHUT OFF. SAFETY CONTROLS: THE CONTROL SYSTEM WILL MONITOR ALL ROTOR BEARING TEMPERATURES INCLUDING GEAR BOX BEARINGS AND MOTOR WINDING TEMPERATURES TO INDICATE TO THE OPERATOR IF UNUSUAL HEATING OF THESE COMPONENTS IS OCCURRING. IF THE TEMPERATURES EXCEED A VALUE OF 82 °C IT WILL ANNUNCIATE AN ALARM TO INDICATE SPECIFICALLY WHAT IS WRONG WITH THE CHILLER AND THE TEMPERATURE VALUES SENSED. MOTOR WINDING TEMPERATURE SENSORS MUST BE IMBEDDED WITHIN THE MOTOR WINDINGS PROPER BY THE MANUFACTURER AT THE TIME OF ASSEMBLY. A LATCHING DIAGNOSTIC SHALL OCCUR IF THE TEMPERATURE OF THE MOTOR WINDING REACHES A TEMPERATURE OF 265 °F. RETROFIT SENSORS ARE NOT ALLOWED.

EVAPORATOR AND CONDENSER REFRIGERANT TEMPERATURES: THE CONTROL SYSTEM WILL MONITOR THESE POINTS IN ORDER TO DETERMINE THE APPROACH TEMPERATURE BETWEEN THE WATER AND THE REFRIGERANT FOR AN INDICATION OF HEAT EXCHANGER EFFICIENCY.

DIAGNOSTICS: THE CONTROL SYSTEM WILL MONITOR DIAGNOSTIC POINTS WITHIN THE CHILLER ITSELF TO DETERMINE IF ANY PROBLEMS EXISTS IN ITS OPERATION. THESE SHALL INCLUDE: HIGH AND LOW CHILLED WATER TEMPERATURES, EVAPORATOR AND CONDENSER REFRIGERANT TEMPERATURES, PHASE IMBALANCE, REVERSAL, LOSS, AND HIGH AND LOW AMPERAGES, MOTOR WINDING TEMPERATURES, BEARING TEMPERATURES, AND GENERAL SENSOR FAILURE.

THE CHILLER CONTROL SYSTEM SHALL RESET THE CHILLED WATER LEAVING TEMPERATURE AS A FUNCTION OF THE OUTSIDE AIR TEMPERATURES. THE FORMULA FOR DETERMINING THE RESET SCHEDULE SHALL BE AS FOLLOWS: THE CHW SUPPLY TEMPERATURE SHALL BE 6.7 °C WHEN OA TEMPERATURE IS 25.6 °C AND ABOVE, 12.2 °C WHEN THE OUTDOOR AIR IS 15.6 °F AND BELOW, AND RAMPED LINEARLY BETWEEN 6.7 °F AND 14.4 °F AT OA TEMPERATURES BETWEEN 15.6 °F AND 25.6 °C.

MECHANICAL SEQUENCE OF OPERATIONS



SOLICITATION NO.:

CONTRACT NO.:

W912HV-17-D-0003
SCALE:

AS SHOWN

TSANG
DESIGNED BY.

DRAWN BY:

HECKED BY:

SUBMITTED BY: CRABTREE

CRABTREE	
ZE:	

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FY20 MCA PN77833 VEHICLE MAINTENANCE SHOP

MECHANICAL SEQUENCE OF OPERATIONS

SHEET ID

MI-707

SINGLE ZONE WITH HEATING AND DX COILS

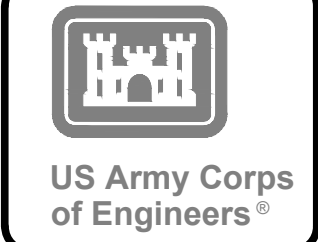
FUNCTION	NAME	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING			PRIMARY POINT INFORMATION	NETWORK DATA EXCHANGE		OVERRIDES	TREND OBJECT LOCATION (L OR R), TYPE AND INSTANCE NUMBER (PROVIDE DDC ID FOR REMOTE IN NOTES)	ALARM INFORMATION			CONFIGURATION INFORMATION
												ALARM CONDITION	ALARM PRIORITY	M&C ROUTING		DATA FROM	DATA TO			ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN NOTES)	NOTIFICATION CLASS OBJECT INSTANCE NUMBER	EVENT ENROLLMENT OBJECT INSTANCE NUMBER	
PROOFS & SAFETIES	SF-S	SUPPLY FAN STATUS	<_>		ON/OFF	BI			V	V	~	SUPPLY FAN PROOF FAILED	CRIT	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	HTG-DA-T-LL	HEATING COIL DISCHARGE TEMPERATURE LOW LIMIT SWITCH	<_>	4 DEG C	ALMNORMAL	BI		H	~	~	~	ALM	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	RST-BUT	SYSTEM RESET BUTTON (FOR SAFETIES)	<_>			BI			VO	~	~				<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
START/STOP	BLDG-T	BUILDING TEMPERATURE (NIGHT STAT)	<_>		<_>	AI			V	~	~	BLDG-T IS LESS THAN BLDG-T-LL	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	BLDG-T-LL	BUILDING TEMPERATURE LOW LIMIT SETPOINT		6 DEG C				C	VO	~	~				<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	SYS-OCC	OCCUPANCY INPUT (FROM SYSTEM SCHEDULER)	<_>		<_>	NET-IN			V	V	~				<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	SF-SS	SUPPLY FAN START/STOP	<_>		ON/OFF	BO	~		VO	V	~				<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	UNIT STATUS	UNIT STATUS (HTG AND/OR CLG REQUEST) (SEE NOTE)	<_>		<HVAC_COOL/HVAC_HEAT/HVAC_OF F>	NET-OUT			V	V	~	~	CRIT	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
SUPPLY FAN CAPACITY CONTROL	SF-C	SUPPLY FAN COMMAND	<_>		0-100%	AO	~		VO	V	X	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	SF-C-MIN	SUPPLY FAN COMMAND MINIMUM		25%				C							<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
MINIMUM OUTSIDE AIRFLOW	MINOA-D-C	MINIMUM OUTSIDE AIR DAMPER COMMAND	<_>		OPEN/CLOSED	BO	~		VO	~	~	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	MINOA-F	MINIMUM OUTSIDE AIR FLOW SETTING		10 LPS				C							<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
ZONE TEMPERATURE CONTROL: MA DAMPERS HEATING COIL COOLING COIL	OA-T	OUTSIDE AIR TEMPERATURE	<_>		<_>	AI	~		V	V	~	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	MA-D-C	MIXED AIR DAMPER COMMAND	<_>		0-100% OPEN	AO	~		VO	V	X	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	MIXED AIR PID LOOP SETTINGS	PROPORTIONAL CONSTANT	<_>	<_>				C							<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
		INTEGRAL CONSTANT	<_>	<_>				C							<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
		BIAS	<_>	<_>				C							<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
		MAXIMUM OUTPUT	<_>	<_>				C							<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	ZN-T	ZONE TEMPERATURE	<_>		<_>	AI			V	V	X	** ZN-T IS MORE THAN 3 DEG C ABOVE OR BELOW ZN-T-SP	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
		ZONE TEMPERATURE SETPOINT		OCC ADJ	6-22 DEG C	AI		C	VO	V	~	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
		DX UNIT COMMAND	<_>		ON/OFF	BO	~		VO	V	X	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	ZONE TEMPERATURE AND HUMIDITY CONTROL	ZN-T	ZONE TEMPERATURE	<_>		<_>	AI		V	V	X	** ZN-T IS MORE THAN 3 DEG C ABOVE OR BELOW ZN-T-SP	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
		ZN-T-SP	ZONE TEMPERATURE SETPOINT		OCC ADJ	6-22 DEG C	AI	C	VO	V	X	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
		ZN-RH	ZONE RELATIVE HUMIDITY	<_>		<0 - 100 % RH>	AI		V	V	X	** ZN-RH IS MORE THAN 10% ABOVE OR BELOW ZN-RH-SP	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
		ZN-RH-SP	ZONE RELATIVE HUMIDITY SETPOINT			<_>	AI	C	~	~	X	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
		SARH	SUPPLY AIR RELATIVE HUMIDITY	<_>		<0 - 100 % RH>	AI		V	V	X	SA-RH IS ABOVE 90% RH	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
		SARH-SP	SUPPLY AIR RELATIVE HUMIDITY SETPOINT		40% RH	<_>	AI	C	~	~	~	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
OTHER POINTS	SA-T	SUPPLY AIR TEMPERATURE	<_>		<_>	AI			V		~	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	SA-T-SP	SUPPLY AIR TEMPERATURE SET POINT		RESET	6-22 DEG C			C	V	~	~	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	MA-FLT-P-HL	MIXED AIR FILTER PRESSURE HIGH LIMIT SWITCH	<_>	<_>	ALMNORMAL	BI		H	V	~	~	ALM	CRIT	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
OPTIONAL DCV POINTS	MA-D-C-MIN1	MINIMUM DAMPER COMMAND 1 (DESIGN MIN OUTSIDE AIR)		10 LPS				C	VO	V	~				<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>

- Notes:
- 1) THE CONTRACTOR SHALL COMPLETE THE POINTS SCHEDULE AS INDICATED IN THE SPECIFICATION AND IN ACCORDANCE WITH THE POINTS SCHEDULE INSTRUCTIONS DRAWING.
 - 2) UNIT MANUFACTURERS PROOFS AND SAFETIES: THE CONTRACTOR SHALL SHOW EACH PROOF AND SAFETY AS A SEPARATE ROW.
 - 3) UNIT STATUS: SERVES AS A MONITORED POINT AT THE M&C SOFTWARE (FRONT-END) AND AS A HEATING/COOLING REQUEST TO THE BOILER, HEAT EXCHANGER, AND/OR CHILLER SERVING THIS SYSTEM.
 - 4) DDC HWID FOR DEVICE CONTAINING TREND OBJECT FOR REMOTE TRENDING FOR THIS SYSTEM <_>

DOMESTIC HOT WATER HEATER

FUNCTION	NAME	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING			PRIMARY POINT INFORMATION	NETWORK DATA EXCHANGE		OVERRIDES	TREND OBJECT LOCATION (L OR R), TYPE AND INSTANCE NUMBER (PROVIDE DDC ID FOR REMOTE IN NOTES)	ALARM INFORMATION			CONFIGURATION INFORMATION
												ALARM CONDITION	ALARM PRIORITY	M&C ROUTING		DATA FROM	DATA TO			ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN NOTES)	NOTIFICATION CLASS OBJECT INSTANCE NUMBER	EVENT ENROLLMENT OBJECT INSTANCE NUMBER	
PROOFS & SAFETIES	HW-PMP-S	HOT WATER PUMP STATUS	<_>		ON/OFF	BI			V	V	~	~	CRIT	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
															<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
START/STOP	SYS-ENA	SYSTEM ENABLE	<_>		ENABLE/DISABLE	NET-IN			VO	~	~	~	CRIT	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	DHW-PMP-SS	DOMESTIC HOT WATER PUMP START/STOP	<_>		START/STOP	BO	~		VO	V	~	~	CRIT	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	UNIT STATUS	HEATING STATUS	<_>		ON/OFF	NET-OUT			V	~	~	~	CRIT	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
OTHER POINTS	DHWS-T-SP	DOMESTIC HOT WATER SUPPLY TEMPERATURE SETPOINT		60 DEG C	<_>			C	VO	V	X	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	DHWR-T	DOMESTIC HOT WATER RETURN TEMPERATURE	<_>		<_>	AI			V	V	X	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	DHW-TANK-T-SP	DOMESTIC HOT WATER TANK TEMPERATURE SETPOINT		60 DEG C	<_>			C	VO	V	X	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	DHW-TANK-T	DOMESTIC HOT WATER TANK TEMPERATURE	<_>		<_>	AI			V	V	X	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
	BLR-S	HEATER STATUS (STATE)	<_>		START/STOP	BI			V	V	X	~	INFO	[]	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>	<_>
													CRIT	[]									

- Notes:
- 1) THE CONTRACTOR SHALL COMPLETE THE POINTS SCHEDULE AS INDICATED IN THE SPECIFICATION AND IN ACCORDANCE WITH THE POINTS SCHEDULE INSTRUCTIONS DRAWING.
 - 2) UNIT MANUFACTURERS PROOFS AND SAFETIES: THE CONTRACTOR SHALL SHOW EACH PROOF AND SAFETY AS A SEPARATE ROW.
 - 3) UNIT STATUS: SERVES AS A MONITORED POINT AT THE M&C SOFTWARE (FRONT-END) AND AS A HEATING/COOLING REQUEST TO THE BOILER, HEAT EXCHANGER, AND/OR CHILLER SERVING THIS SYSTEM.
 - 4) DDC HWID FOR DEVICE CONTAINING TREND OBJECT FOR REMOTE TRENDING FOR THIS SYSTEM <_>



ISSUE DATE:	DATE
DESIGNED BY:	MARK
DRAWN BY:	DESCRIPTION
CHECKED BY:	
SUBMITTED BY:	
SCALE:	
AS SHOWN	

US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA AREA OFFICE APO AP 96376-5132	DESIGN PARTNERS INCORPORATED 1580 MAKALOA STREET, SUITE 1100 HONOLULU, HAWAII 96814
ISSUED BY: O CRAWFORD	SCALE: AS SHOWN
CONTRACT NO.: W01HW-17-D-0003	SIZE: ANSI D
SOLICITATION NO.:	ISSUE DATE:

APC AP 96343-5010 FY20 MCA PN77833 VEHICLE MAINTENANCE SHOP	DDC POINT LIST
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SHEET ID MI-709

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INTERMEDIATE DESIGN

FUNCTION	NAME	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	ID TYPE	HOA REQ'D	CONFRG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARM CONDITION	ALARM PRIORITY	M&C ROUTING	OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	DATA FROM	DATA TO	OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	LOCATION (L OR R), TYPE AND INSTANCE NUMBER (PROVIDE DDC ID FOR REMOTE IN NOTES)	ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN NOTES)	NOTIFICATION CLASS OBJECT INSTANCE NUMBER	EVENT ENROLLMENT OBJECT INSTANCE NUMBER	CONFIGURATION INFORMATION	
PROOFS & SAFETIES		<UNIT MANUFACTURER'S SAFETIES> (SEE NOTES)		<>	ALMNORMAL	<BI>	~	H	~	~	~	ALM	CRIT	[]	<>	<>	<>	<>	<>	<>	<>	<>	<>	
		<UNIT MANUFACTURER'S SAFETIES> (SEE NOTES)		<>	ALMNORMAL	<BI>	~	H	~	~	~	ALM	CRIT	[]	<>	<>	<>	<>	<>	<>	<>	<>	<>	
START/STOP	SYS-OCC	OCCUPANCY INPUT (FROM SYSTEM SCHEDULER)	<>		<>	NET-IN			VO	~	~	~	INFO	[]	MSV<>	<>	<>	<>	<>	<>	<>	<>	<>	
	ZN-OCC	ZONE OCCUPANCY INPUT (OCCUPANCY SENSOR)	<>		<>	BI			~	~	~	~	INFO	[]	MSV<>	<>	<>	<>	<>	<>	<>	<>	<>	
	EFF-OCC	EFFECTIVE OCCUPANCY	<>		<>	NET-OUT			V	~	~	~	INFO	[]	MSV<>	<>	<>	<>	<>	<>	<>	<>	<>	
	UNIT STATUS	UNIT STATUS (HTG AND/OR CLG REQUEST) (SEE NOTE)	<>		HVAC_COOL/ HVAC_HEAT/ HVAC_OFF	NET-OUT			V	~	~	~	CRIT	[]	<>	<>	<>	<>	<>	<>	<>	<>	<>	
ZONE TEMPERATURE CONTROL	ZN-T	ZONE TEMPERATURE	<>		<>	AI			V	~	X	** ZN-T IS MORE THAN 3 DEG C ABOVE OR BELOW ZN-T-SP	INFO	[]	<>	<>	<>	<>	<>	<>	<>	<>	<>	
	ZN-T-SP	ZONE TEMPERATURE SETPOINT		OCC ADJ.	20-25 DEG C	AI		C	VO	~	~	~			<>	<>	<>	<>	<>	<>	<>	<>	<>	
	ZN-T-SP-UNOCC	ZONE TEMPERATURE SETPOINT FOR UNOCCUPIED MODE	<>	28 DEG C	<>	AI		C	~	~	~	~	INFO	[]	<>	<>	<>	<>	<>	<>	<>	<>	<>	
	VAV-SA-F	VAV SUPPLY AIR FLOW	<>	SEE SCHED	SEE MANUFACTURER MANUAL	AI			V	~	~	~	INFO	[]	<>	<>	<>	<>	<>	<>	<>	<>	<>	
	VAV-SA-F-SP	VAV SUPPLY AIR FLOW SETPOINT		RESET SCHED	SEE MANUFACTURER MANUAL			C	~	~	~	~			<>	<>	<>	<>	<>				<>	
	VAV-D-C	VAV DAMPER COMMAND	<>		0-100% OPEN	AO	~		V	~	X	~	INFO	[]	<>	<>	<>	<>	<>	<>	<>	<>	<>	
	VAV-FAN-SS	VAV FAN START/STOP	<>		ON/OFF	BO	~		V	~	~	~	CRIT	[]	<>	<>	<>	<>	<>	<>	<>	<>	<>	
		ZONE TEMPERATURE SETPOINT DEADBAND		RESET SCHED	SEE MANUFACTURER MANUAL			C	~	~	~	~			<>	<>	<>	<>	<>				<>	
	VAV DAMPER PID LOOP SETTINGS	PROPORTIONAL CONSTANT	<>	<>					C															<>
		INTEGRAL CONSTANT	<>	<>					C															<>
		BIAS	<>	<>					C															<>
		MAXIMUM OUTPUT	<>	<>					C															<>
		MINIMUM OUTPUT	<>	<>					C															<>
		<->	<>	<>					C															
	VAV VALVE PID LOOP SETTINGS	PROPORTIONAL CONSTANT	<>	<>					C															<>
		INTEGRAL CONSTANT	<>	<>					C															<>
		BIAS	<>	<>					C															<>
		MAXIMUM OUTPUT	<>	<>					C															<>
MINIMUM OUTPUT		<>	<>					C															<>	
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INTERMEDIATE DESIGN

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US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT CHINA APO AF 96376-5132	DRAWN BY: I. TSANG I. TSANG	SOLICITATION NO.: 1550
DESIGN PARTNERS INCORPORATED 1350 KALANIAN'OLE AVENUE HONOLULU, HAWAII 96814	CONTRACT NO. W01HN-11-D-0003 O. CRABTREE SUBMITTED BY: O. CRABTREE SCALE: AS SHOWN	ISSUE DATE: 11/1/81

SHEET ID
MI-712

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DEDICATED OUTSIDE AIR UNIT

FUNCTION	NAME	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFRG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING	PRIMARY POINT INFORMATION	NETWORK DATA EXCHANGE	OVERRIDES	TREND OBJECT LOCATION (L OR R), TYPE AND INSTANCE NUMBER (PROVIDE DDC ID FOR REMOTE IN NOTES)	ALARM INFORMATION	CONFIGURATION INFORMATION					
												ALARM CONDITION	ALARM PRIORITY	M&C ROUTING	OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	DATA FROM	DATA TO	OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN NOTES)	NOTIFICATION CLASS OBJECT INSTANCE NUMBER	EVENT ENROLLMENT OBJECT INSTANCE NUMBER		
PROOFS & SAFETIES	SF-S	SUPPLY FAN STATUS	<_>		ON/OFF	BI			V	V		SUPPLY FAN PROOF FAILED	CRIT	☐	<_>	<_>	<_>		☐	☐	☐		
	SA-SMK	SUPPLY AIR SMOKE	<_>		ALMNORMAL	BI		H				ALM	CRIT	☐	<_>	<_>	<_>		☐	☐	☐		
	HTG-DAT-LL	HEATING COIL DISCHARGE TEMPERATURE LOW LIMIT SWITCH	<_>	4 DEG C	ALMNORMAL	BI		H				ALM	CRIT	☐	<_>	<_>	<_>		☐	☐	☐		
	RST-BUT	SYSTEM RESET BUTTON (FOR SAFETIES)				BI		H							<_>	<_>	<_>						
	EMC-BUT	EMERGENCY STOP BUTTON	<_>			BI							ALM	CRIT	☐	<_>	<_>	<_>	☐				
OA-FLT-P-HL	AIR FILTER PRESSURE HIGH LIMIT SWITCH	<_>	<_>	ALMNORMAL	BI		H		V			ALM	INFO	☐	<_>	<_>	<_>		☐	☐	☐		
START/STOP	SYS-OCC	OCCUPANCY INPUT (FROM SYSTEM SCHEDULER)	<_>		<_>	NET-IN			V	V					MSV <_>	<_>	<_>						
	SF-SS	SUPPLY FAN START/STOP	<_>		ON/OFF	BO	~		VO	V						<_>	<_>	<_>					
	UNIT STATUS	UNIT STATUS (SEE NOTE)	<_>		ON/OFF	NET-OUT			V							<_>	<_>	<_>					
ATFP	LLO-D-2P	ATFP LOW LEAKAGE OUTSIDE AIR DAMPER COMMAND (2-POS)	<_>		OPEN/CLOSED	BO	~		V						<_>	-	-	-		☐	☐	☐	
	LLE-D-2P	ATFP LOW LEAKAGE EXHAUST AIR DAMPER COMMAND (2-POS)	<_>		OPEN/CLOSED	BO	~		V						<_>	-	-	-		☐	☐	☐	
	ATFP-SS	ATFP EMERGENCY SYSTEM STOP	<_>		ON/OFF	BI	~		V			DE-ENERGIZE ENTIRE OUTSIDE AIR SYSTEM AND CLOSE LOW LEAKAGE DAMPERS.	CRIT	☐	<_>	-	-	-		☐	☐	☐	
FAN CAPACITY CONTROL	SA-F	SUPPLY AIR FLOW	<_>		<_>	AI									<_>	<_>	<_>		☐	☐	☐		
	EA-F	EXHAUST AIR FLOW	<_>		<_>	AI									<_>	<_>	<_>		☐	☐	☐		
SUPPLY AIR TEMPERATURE	SA-T	SUPPLY AIR TEMPERATURE	<_>		<_>	AI			V		X	*SA-T MORE THAN 10 DEG C ABOVE OR BELOW SETPOINT	INFO	☐	<_>	<_>	<_>		<_>	☐	☐	☐	
	SA-T-SP	SUPPLY AIR TEMPERATURE SETPOINT		13 DEG C	<_>			C	VO	VO					<_>	<_>	<_>		<_>				
	CLG-V-C	COOLING VALVE COMMAND	<_>		0-100% OPEN	AO	~		VO	V					<_>	<_>	<_>		<_>				
	HTG-V-C	HEATING VALVE PID LOOP SETTINGS	<_>	<_>	0-100% OPEN	AO	~		VO	V					<_>	<_>	<_>		<_>				
	HTG-V-C	HEATING VALVE PID LOOP SETTINGS	<_>	<_>	0-100% OPEN	AO	~		C						<_>	<_>	<_>		<_>				
OTHER POINTS	OA-T	OUTSIDE AIR TEMPERATURE	<_>		<_>	AI			V			~		~	<_>	<_>	<_>		☐	☐	☐		
ENERGY RECOVERY VENTILATOR (ERV) POINTS	OA2-T	OUTSIDE AIR TEMPERATURE 2 (DISCHARGE OF HEAT WHEEL)	<_>		<_>	AI			V			~		~	<_>	<_>	<_>		☐	☐	☐		
	EA2-T	EXHAUST AIR TEMPERATURE 2 (DISCHARGE OF HEAT WHEEL)	<_>		<_>	AI			V			~		~	<_>	<_>	<_>		☐	☐	☐		
	ERV-FLT-P-HL	ERV AIR FILTER PRESSURE HIGH LIMIT SWITCH	<_>	<_>	ALMNORMAL	BI		H		V		ALM	INFO	☐	<_>	<_>	<_>		☐	☐	☐		

Notes:
1) THE CONTRACTOR SHALL COMPLETE THE POINTS SCHEDULE AS INDICATED IN THE SPECIFICATION AND IN ACCORDANCE WITH THE POINTS SCHEDULE INSTRUCTIONS DRAWING.
2) UNIT MANUFACTURERS PROOFS AND SAFETIES. THE CONTRACTOR SHALL SHOW EACH PROOF AND SAFETY AS A SEPARATE ROW.
3) UNIT STATUS: SERVES AS A MONITORED POINT AT THE M&C SOFTWARE (FRONT-END) AND AS A HEATING/COOLING REQUEST TO THE BOILER, HEAT EXCHANGER, AND/OR CHILLER SERVING THIS SYSTEM.
4) DDC HW ID FOR DEVICE CONTAINING TREND OBJECT FOR REMOTE TRENDING FOR THIS SYSTEM <_>

EXHAUST FANS

FUNCTION	NAME	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFRG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING	PRIMARY POINT INFORMATION	NETWORK DATA EXCHANGE	OVERRIDES	ALARM INFORMATION	CONFIGURATION INFORMATION					
												ALARM CONDITION	ALARM PRIORITY	M&C ROUTING	OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	DATA FROM	DATA TO	OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN NOTES)	NOTIFICATION CLASS OBJECT INSTANCE NUMBER		
PROOFS & SAFETIES	EF-S	EXHAUST FAN STATUS	<_>	-	ON/OFF	BI	-	-	V	V	-	EXHAUST FAN PROOF FAILED	CRIT	☐	<_>	<_>	<_>	-	<_>	<_>	-	
	RST-BUT	SYSTEM RESET BUTTON (FOR SAFETIES)	-	-	-	BI	-	H	-	-	-	-	-	-	<_>	<_>	<_>	-	-	-	-	
	EMC-BUT	EMERGENCY STOP BUTTON	<_>			BI							ALM	CRIT	☐	<_>	<_>	<_>		☐		☐
ATFP	LLO-D-2P	ATFP LOW LEAKAGE OUTSIDE AIR DAMPER COMMAND (2-POS)	<_>		OPEN/CLOSED	BO	~		V						<_>	<_>	<_>	-	-	-	-	
	LLE-D-2P	ATFP LOW LEAKAGE EXHAUST AIR DAMPER COMMAND (2-POS)	<_>		OPEN/CLOSED	BO	~		V						<_>	<_>	<_>	-	-	-	-	
	ATFP-SS	ATFP EMERGENCY SYSTEM STOP	<_>		ON/OFF	BI	~		V			DE-ENERGIZE ENTIRE OUTSIDE AIR SYSTEM AND CLOSE LOW LEAKAGE	CRIT	☐	<_>	<_>	<_>	-	-	-	-	
START/STOP	SYS-OCC	OCCUPANCY INPUT (FROM SYSTEM SCHEDULER)	<_>	-	<_>	NET-IN	-	-	V	V	-	-	-	-	MSV <_>	<_>	<_>	-	-	-	-	
	EF-SS	EXHAUST FAN START/STOP	<_>	-	ON/OFF	BO	-	-	VO	V	-	-	-	-	<_>	<_>	<_>	<_>	-	-	-	-
FAN CAPACITY CONTROL	EA-F	EXHAUST AIR FLOW	<_>	-	<_>	AI	-	-			X	-	-	-	<_>	<_>	<_>	-	-	-	-	
	EA-T	EXHAUST AIR TEMPERATURE	<_>	-	<_>	AI	-	-	V	-	-	-	-	-	<_>	<_>	<_>	-	-	-	-	

Notes:
1) THE CONTRACTOR SHALL COMPLETE THE POINTS SCHEDULE AS INDICATED IN THE SPECIFICATION AND IN ACCORDANCE WITH THE POINTS SCHEDULE INSTRUCTIONS DRAWING.
2) UNIT MANUFACTURERS PROOFS AND SAFETIES. THE CONTRACTOR SHALL SHOW EACH PROOF AND SAFETY AS A SEPARATE ROW.
3) UNIT STATUS: SERVES AS A MONITORED POINT AT THE M&C SOFTWARE (FRONT-END) AND AS A HEATING/COOLING REQUEST TO THE BOILER, HEAT EXCHANGER, AND/OR CHILLER SERVING THIS SYSTEM.
4) DDC HW ID FOR DEVICE CONTAINING TREND OBJECT FOR REMOTE TRENDING FOR THIS SYSTEM <_>

US Army Corps of Engineers

DESIGNED BY:
L TSANG
DRAWN BY:
JAS
CHECKED BY:
O CRABTREE
SUBMITTED BY:
O CRABTREE
SCALE:
AS SHOWN
SIZE:
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US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA AREA OFFICE
APO AP 96376-5132
DESIGN PARTNERS INCORPORATED
1580 MAKALOA STREET, SUITE 1100
HONOLULU, HAWAII 96814

APO AP 96343-5010
FY20 MCA PN77833 VEHICLE MAINTENANCE SHOP
DDC POINT LIST

SHEET ID
MI-713

INTERMEDIATE DESIGN

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- APPROXIMATE AND BEFORE INSTALLING, STUDY THE ARCHITECTURAL, STRUCTURAL, AND MECHANICAL DETAILS, THEN MAKE INSTALLATION IN THE MOST LOGICAL MANNER. CIRCUIT ROUTING IS TYPICAL AND MAY BE VARIED IN ANY MANNER. ANY PIECE OF EQUIPMENT/DEVICE MAY BE RELOCATED WITHIN 10' BEFORE INSTALLATION AT THE DIRECTION OF THE CONTRACTING OFFICER WITHOUT ADDITIONAL CHARGE TO THE PROJECT.
7. SHOULD PROJECT CONDITIONS REQUIRE REARRANGEMENT OF THE PROJECT'S WORK, THE CONTRACTOR SHALL MARK SUCH CHANGES ON THE AS-BUILT DRAWINGS. IF THESE CHANGES REQUIRE AN ALTERNATE METHOD TO THOSE SPECIFIED IN THE CONTRACT DOCUMENTS, THE CONTRACTOR SHALL SUBMIT DRAWINGS TO REFLECT THE PROPOSED ALTERNATE METHODS TO THE CONTRACTING OFFICER FOR REVIEW AND APPROVAL. THE CONTRACTOR SHALL NOT PROCEED UNTIL APPROVAL IS OBTAINED. REARRANGEMENT OF WORK FOR THE PURPOSE OF COORDINATION SHALL NOT BE CONSIDERED AN ITEM FOR EXTRA COST.
 8. THE EXISTING ELECTRICAL, TELECOM, FIRE ALARM, AND OTHER ELECTRICALLY-RELATED SYSTEMS IN AREAS ADJACENT TO, OUTSIDE OF, AND/OR OTHERWISE PASSING THROUGH THE PROJECT LIMITS, MUST REMAIN OPERATIONAL DURING THE CONSTRUCTION PERIOD AND POST-CONSTRUCTION. THE CONTRACTOR SHALL EXERCISE DUE CARE AND CAUTION WHEN WORKING NEAR ANY EXISTING EQUIPMENT, DEVICES, OR CABLING/CIRCUITING. PROVIDE NEW JUNCTION BOXES, CONDUITS, WIRING, AND THE LABOR REQUIRED TO FACILITATE THE REQUIRED OPERATIONAL CONTINUITY. BOXES, CONDUITS, AND WIRING SHALL BE IN ACCORDANCE WITH THE NEC. ANY DAMAGE TO THE EXISTING EQUIPMENT, DEVICES, OR CABLING/CIRCUITING RESULTING FROM THE CONTRACTOR'S OPERATIONS SHALL BE IMMEDIATELY REPAIRED OR OTHERWISE RESTORED TO ITS ORIGINAL WORKING CONDITION AT NO ADDITIONAL COST TO THE PROJECT.
 9. THE ELECTRICAL DRAWINGS ARE BASED ON PROPOSED EQUIPMENT. VERIFY ALL SYSTEM REQUIREMENTS (ELECTRICAL, MECHANICAL, FIRE ALARM, SPECIALTY SYSTEMS, ETC.) WITH THE SELECTED SYSTEM'S MANUFACTURER OR AUTHORIZED REPRESENTATIVE PRIOR TO COMMENCING WITH ANY WORK. COORDINATE RATINGS OF OVERCURRENT PROTECTION DEVICES, DISCONNECT SWITCHES, CONDUIT, AND WIRING TO MATCH THE ACTUAL EQUIPMENT SUPPLIED FOR THE PROJECT. CORRECT ALL DISCREPANCIES SO AS TO PROVIDE A COMPLETE AND OPERATIONAL SYSTEM. RECORD CHANGES ON THE AS-BUILT DRAWINGS.
 10. ALL EQUIPMENT AND APPARATUS SHALL BE CAPABLE OF FITTING IN THE SPACES SHOWN WHILE MEETING THE MANUFACTURER'S RECOMMENDED ACCESS REQUIREMENTS AND APPLICABLE CODE REQUIREMENTS. REVIEW ALL SPACES WHERE EQUIPMENT IS TO BE INSTALLED PRIOR TO ORDERING OF EQUIPMENT AND NOTIFY THE CONTRACTING OFFICER OF ANY INADEQUATE CLEARANCES OR CONDITIONS THAT WILL PREVENT THE PROPER INSTALLATION, MAINTENANCE, AND OPERATION OF THE EQUIPMENT.
 11. CONFIRM THE TYPE OF CEILING BEING INSTALLED PRIOR TO ORDERING LUMINAIRES AND TRIMS FOR PROPER COORDINATION. LUMINAIRES INDICATED MAY NOT EXPRESSLY CONFORM TO THE TYPE OF CEILING OR OPENING PROVIDED BY OTHER TRADES.
 12. CONCEAL ALL CONDUIT [WHEREVER REASONABLY POSSIBLE]; EXPOSED CONDUITS ARE PERMITTED ONLY WHERE SPECIFICALLY SHOWN ON THE DRAWINGS. ALL EXPOSED CONDUITS IN FINISHED AREAS SHALL BE INSTALLED IN THE LEAST VISIBLE LOCATIONS. CARE SHALL BE TAKEN TO INSTALL CONDUIT IN THE MOST AESTHETICALLY PLEASING MANNER.
 13. WHERE DEVICES, EQUIPMENT, BOXES, AND OTHER ELECTRICAL MATERIALS ARE INDICATED TO BE OF WEATHERPROOF (WP) CONSTRUCTION, THE INTERCONNECTING RACEWAYS, INCLUDING COUPLINGS AND CONNECTORS, SHALL BE LISTED FOR AND INSTALLED TO PROVIDE A COMPLETE WEATHERPROOF / WET LOCATION INSTALLATION. ALL INCIDENTAL MATERIALS REQUIRED TO COMPLETE THE INSTALLATION SHALL ALSO BE OF WEATHERPROOF / WET LOCATION LISTING.
 14. WIRING DEVICES AND CONDUITS SHALL BE FLUSH MOUNTED, WHEREVER REASONABLY POSSIBLE.
 15. PROVIDE TYPEWRITTEN CIRCUIT DIRECTORIES FOR ALL PANELS, NEW OR MODIFIED, REFLECTING THE CIRCUIT ARRANGEMENTS AS THEY WERE ACTUALLY INSTALLED.
 16. AN ADHESIVE VINYL NAMEPLATE SHALL BE PROVIDED FOR ALL SWITCHES, RECEPTACLES, [MODULAR FURNITURE POWER CONNECTIONS,] [DISCONNECT SWITCHES,] [MOTOR STARTERS,] AND MISCELLANEOUS DEVICES REQUIRING POWER. THE NAMEPLATE SHALL INDICATE THE PANELBOARD SERVING THE DEVICE AND THE CORRESPONDING CIRCUIT ASSIGNMENT. LETTERING SHALL BE A MINIMUM OF 1/4" HIGH. UTILIZE BROTHER "P-TOUCH" LABEL MAKER OR APPROVED SUBSTITUTE.
 17. A GREEN, EQUIPMENT GROUND CONDUCTOR SIZED IN ACCORDANCE WITH THE NEC ARTICLE 250 SHALL BE INSTALLED IN ALL FEEDER AND BRANCH CIRCUITS WHETHER INDICATED ON CONTRACT DRAWINGS OR NOT. INSTALL THIS CONDUCTOR IN ALL RACEWAYS INCLUDING THOSE INSTALLED FOR SWITCH LEGS AND ATTACH TO THE DEVICE, LUMINAIRE, OR EQUIPMENT USING A SUITABLE GROUNDING LUG.
 18. DO NOT USE A COMMON NEUTRAL FOR MULTIPLE BRANCH CIRCUITS INSTALLED IN A COMMON CONDUIT. PROVIDE A DEDICATED NEUTRAL FOR EACH INDIVIDUAL CIRCUIT. WHERE MULTIPLE DEDICATED NEUTRALS ARE INSTALLED IN A COMMON CONDUIT, PROVIDE COLOR CODING OF THE DIFFERENT NEUTRAL CONDUCTORS IN ACCORDANCE WITH THE NEC (WHITE, GRAY, THREE CONTINUOUS WHITE OR GRAY STRIPES, ETC.)
 19. PROVIDE NYLON PULLSTRINGS IN ALL EMPTY CONDUITS UNLESS OTHERWISE INDICATED.
 20. THE TELECOMMUNICATIONS RACEWAY SYSTEM INSTALLATION SHALL COMPLY WITH TIA/EIA AND BICSI STANDARDS UNLESS OTHERWISE NOTED.
 21. CONDUIT BODIES (E.G. LB, LR, ETC.) SHALL NOT BE PERMITTED IN THE TELECOMMUNICATIONS RACEWAY SYSTEMS UNLESS SPECIFICALLY INDICATED TO BE UTILIZED AND LISTED FOR TELECOMMUNICATIONS SYSTEM USE.
 22. PROVIDE INSULATED BUSHINGS AT ALL TELECOMMUNICATIONS CONDUIT TERMINATIONS AT ALL BOXES, BACKBOARDS, AND CONDUIT STUBS.
 23. ALL SURFACE MOUNTED DEVICES SHALL BE INSTALLED UTILIZING FACTORY PAINTED SURFACE MOUNTING ACCESSORIES AND MATCHING DEVICE BOXES FOR THE MOST AESTHETICALLY PLEASING INSTALLATION.
 24. PROVIDE KNOCK-OUT PLUGS FOR ALL UNUSED CONDUIT PENETRATIONS IN BOXES AND ENCLOSURES DUE TO CONDUIT REMOVAL.
 25. PENETRATIONS THROUGH FIRE-RATED WALLS, CEILINGS, AND FLOORS SHALL BE SEALED TO MAINTAIN FIRE RATINGS. UTILIZE 3M CP25, PUTTY 303, OR OTHER SUITABLE UL-LISTED SEALING SYSTEM.
 26. PATCH, REFINISH, AND PAINT ALL PENETRATIONS THROUGH WALLS AND SLABS TO MATCH FINISH OF ADJACENT SURFACES.

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US ARMY CORPS OF ENGINEERS JAPAN DISTRICT, OKINAWA AREA OFFICE USAED-J UNIT 35132 APO, AF 96374-5132	DESIGNED BY:	ISSUE DATE:
	DRAWN BY:	SOLICITATION NO.:
	CHECKED BY:	CONTRACT NO.:
	SUBMITTED BY:	SCALE: AS SHOWN
 SSFM INTERNATIONAL, INC. 50 KONOULU, HAWAII 96817	SIZE: ISO A1	FILE NAME:

SHEET ID
E-001

MOUNTING HEIGHT FROM FLOOR TO		SYMBOL		DESCRIPTION
TOP	ℓ	EXISTING	NEW	
	1500			JUNCTION BOX
				VARIABLE FREQUENCY DRIVE, SEE MECH. SPECIFICATIONS AND DWGS FOR REQUIREMENTS
1100				ENCLOSED CIRCUIT BREAKER. LOCKABLE AND NEMA 4X STAINLESS STEEL ENCLOSURE FOR EXTERIOR USE.
1500				NON-FUSED SAFETY SWITCH, AMPACITY AND VOLTAGE TO MATCH CIRCUITING, LOCKABLE, HP RATED, NEMA 4X STAINLESS STEEL ENCLOSURE FOR EXTERIOR
				MOTOR STARTER. FURNISHED BY MECHANICAL AND INSTALLED BY ELECTRICAL
				PANELBOARD
				ELECTRICAL EQUIPMENT
				RECEPTACLE, DUPLEX, NEMA 5-20R
				RECEPTACLE, DUPLEX, NEMA 5-20R, GFCI; WP = WEATHER PROOF
		-eP-----	-P----- -S-----	PRIMARY UNDERGROUND DUCT; eP/P DENOTES BOTH EXISTING AND NEW DUCTS; S DENOTES SECONDARY DUCT.
			-LV-----	LOW VOLTAGE CONTROL WIRING RACEWAY, 20 MINIMUM CONDUIT, CONDUCTORS AS INDICATED. PROVIDE JUNCTION BOXES AS REQUIRED.
			-----	EXPOSED RACEWAY, PROVIDE PROPER SUPPORT AT REQUIRED SPACING. EXTERIOR INSTALLATION SHALL BE RIGID GALVANIZED CONDUIT.
				CONCEALED CONDUIT IN CEILING OR WALLS, (HASH MARKS INDICATE NUMBER OF CURRENT CARRYING CONDUCTORS AND 1 GROUND CONDUCTOR WITHIN) #12AWG MINIMUM
			----->	FEEDER DESTINATION INDICATOR
				EQUIPMENT MOTOR TERMINATION WITH FLEXIBLE CONDUIT WHIP
				CONDUIT STUB, 25C MINIMUM CONDUIT UNLESS OTHERWISE NOTED
			+1000	DENOTES 1000 ABOVE FINISHED FLOOR FINISHED GRADE
			WP	DENOTES WEATHERPROOF
				NOTE INDICATOR
				DENOTE ITEMS TO BE DEMOLISHED
			3#250mm ² , 38mm ² G (104C)	FEEDER SIZE CALL OUT; THREE 250mm ² CONDUCTORS AND ONE 38mm ² GROUND CONDUCTOR IN 104mm CONDUIT
				610mm x 1220mm LIGHT FIXTURE, MOUNTING AS INDICATED, SEE SHEETS EL610-EL613 FOR FIXTURE SCHEDULE
				1220mm LINEAR LIGHT FIXTURE, MOUNTING AS INDICATED, SEE SHEETS EL610-EL613 FOR FIXTURE SCHEDULE
				LIGHT FIXTURE, TYPE AND MOUNTING AS INDICATED, SEE SHEETS EL610-EL613 FOR FIXTURE SCHEDULE
				1220mm LINEAR WALL MOUNTED LIGHT FIXTURE, SEE SHEETS EL610-EL613 FOR FIXTURE SCHEDULE
				FLUORESCENT STRIP FIXTURE, MOUNTED AS INDICATED, SEE SHEETS EL610-EL613 FOR FIXTURE SCHEDULE
				FLUORESCENT LIGHT FIXTURE, WALL MTD, SEE SHEETS EL610-EL613 FOR FIXTURE SCHEDULE
				EXIT SIGN, WALL MOUNTED, +2500mm AFF (OR 300mm ABOVE DOOR)
				OCCUPANCY SENSOR, CEILING MTD
				LIGHTING SWITCH, TYPE AS INDICATED, CONTROLLING CIRCUIT "a", OR AS NOTED
				LIGHTING SWITCH, 3-WAY, TYPE AS INDICATED, CONTROLLING CIRCUIT "a", OR AS NOTED
				RECEPTACLE, QUAD, NEMA 5-20R
				MANUAL MOTOR STARTER, HP RATED, WITH THERMAL OVERLOAD ELEMENT
				LOW VOLTAGE SWITCH
				OCCUPANCY SENSOR SWITCH
				ELECTRICAL JUNCTION BOX, CEILING MTD, AS INDICATED
				ELECTRICAL JUNCTION BOX, WALL MTD, "XX": DW = DISHWASHER, HD = HAND DRYER, R = RANGE, WD = WASHER/DRYER, OR AS INDICATED
				MECHANICAL EQUIPMENT CONNECTION, AS INDICATED
				ELECTRICAL EQUIPMENT CONNECTION, AS INDICATED

- THE CONTRACTOR SHALL COMPLY WITH THE CONSTRUCTION PRACTICES AND REQUIREMENTS OF THE LATEST EDITION OF THE NATIONAL ELECTRICAL CODE (NFPA 70), NATIONAL ELECTRICAL SAFETY CODE, OR JAPANESE INDUSTRIAL STANDARDS (JIS) AS NOTED IN SPECIFICATIONS, APPLICABLE INSTRUCTIONS OF MANUFACTURERS OF EQUIPMENT AND MATERIALS SUPPLIED FOR THE PROJECT, AND ALL ORDINANCES, RULES AND POLICIES OF THE FEDERAL, COUNTRY IN WHICH THE WORK IS TO BE PERFORMED.
2. PROVIDE AND INSTALL ALL JUNCTION AND PULLBOXES REQUIRED FOR THE INSTALLATION OF ELECTRICAL DEVICES AND EQUIPMENT, WHETHER OR NOT SPECIFICALLY INDICATED ON THE PLANS. SIZE BOXES PER THE NATIONAL ELECTRICAL CODE.
3. COORDINATE WITH OTHER TRADES AND CONFIRM THE EXACT EQUIPMENT LOCATIONS AND THE LOCATIONS AND MOUNTING HEIGHTS OF ALL POWER AND TELEPHONE DEVICES BEFORE ROUGHING IN.
4. SEAL ALL PENETRATIONS THROUGH FIRE RATED WALLS AND SLABS USING A U.L. LISTED FIRE RATED SEALANT TO MAINTAIN THE INTEGRITY OF THE FIRE RATING. SEE ARCHITECTURAL PLAN FOR FIRE WALL REQUIREMENTS.
5. OBTAIN APPROVAL FROM THE CONTRACTING OFFICER/OFFICER IN CHARGE, BEFORE MAKING ANY PENETRATIONS THROUGH STRUCTURAL MEMBERS OR FIRE RATED WALLS AND SLABS.
6. ALL EQUIPMENT SHALL BE CAPABLE OF FITTING IN THE SPACES LOCATED WHILE MEETING THE MANUFACTURER'S RECOMMENDED ACCESS REQUIREMENTS. REVIEW ALL SPACES WHERE EQUIPMENT IS TO BE INSTALLED PRIOR TO ORDERING OF EQUIPMENT AND NOTIFY THE CONTRACTING OFFICER OF ANY INADEQUATE CLEARANCES OR CONDITIONS THAT WILL PREVENT THE PROPER INSTALLATION, MAINTENANCE, AND OPERATIONS OF THE EQUIPMENT.
7. COORDINATE WITH OTHER TRADES PRIOR TO COMMENCING AND DURING CONSTRUCTION. OVERLAY PLANS AND CHECK FOR ANY DISCREPANCIES OR CONFLICTS WITH OTHER TRADES.
8. VERIFY AND COORDINATE ALL ROOF, WALL AND FLOOR PENETRATIONS WITH THE STRUCTURAL AND ARCHITECTURAL DRAWINGS PRIOR TO THE START OF CONSTRUCTION.
9. PATCH, PRIME AND PAINT ALL EXPOSED ELECTRICAL MATERIAL, INCLUDING CONDUIT, JUNCTION BOXES, PANELS, AND DEVICES TO MATCH ADJACENT SURFACES OR AS INDICATED.
10. PROVIDE ACCESS PANELS TO ALL CONCEALED TRANSFORMERS, DEVICES, JUNCTION BOXES AND EQUIPMENT. COORDINATE THE LOCATION OF ACCESS PANELS TO INSURE THAT THE EQUIPMENT CAN BE MAINTAINED ADEQUATELY.
11. SEISMICALLY BRACE ALL EQUIPMENT AND CONDUIT IN ACCORDANCE WITH SEISMIC ZONE 4 REQUIREMENTS.
12. COORDINATE WITH THE USER FOR ANY OUTAGES ON THE ELECTRICAL SERVICE IN THE PROJECT AREA TO MINIMIZE DISRUPTION TO OTHER FACILITIES. THE CONTRACTOR SHALL NOTIFY AND REQUEST THE OUTAGE THRU THE CONTRACTING OFFICER IN WRITING 60 CALENDAR DAYS PRIOR TO THE PROPOSED OUTAGE.
13. PROVIDE PROPER CONDUIT EXPANSION JOINT FITTINGS WHERE CONDUIT RUNS TRAVERSE AN EXPANSION JOINT.
14. OUTLET BOXES IN FIRE RATED WALLS SHALL HAVE A MINIMUM SEPARATION OF 610MM FROM ANY OUTLET BOXES ON THE OPPOSITE SIDE OF SAME WALL, UNLESS ALLOWED OTHERWISE BY BUILDING CODES.
15. PROVIDE A VAPOR-TIGHT AND WATERTIGHT SEAL AROUND CONDUCTORS IN EACH CONDUIT ENTERING A BUILDING FROM THE EXTERIOR. LOCATE THE SEAL AT THE FIRST CONDUIT TERMINATION IN THE BUILDING.

	US SYSTEM			JAPANESE SYSTEM		
AMPACITY	VOLTAGE	CONDUCTOR SIZE (CU)	CONDUCTOR TYPE	VOLTAGE	CONDUCTOR SIZE (CU)	CONDUCTOR TYPE
26A	600V	12 AWG	THWN/XHHW	600V	2.0MM	EM-IE
30A	600V	10 AWG	THWN/XHHW	600V	3.5MM ²	EM-IE
37A	600V	8 AWG	THWN/XHHW	600V	5.5MM ²	EM-IE
46A	600V	8 AWG	THWN/XHHW	600V	8.0MM ²	EM-IE
66A	600V	4 AWG	THWN/XHHW	600V	14MM ²	EM-IE
86A	600V	2 AWG	THWN/XHHW	600V	22MM ²	EM-IE
122A	600V	1 AWG	THWN/XHHW	600V	38MM ²	EM-IE
164A	600V	2/0 AWG	THWN/XHHW	600V	60MM ²	EM-IE
225A	600V	2/0 AWG	THWN/XHHW	600V	100MM ²	EM-IE
298A	600V	350 kcmil	THWN/XHHW	600V	150MM ²	EM-IE
354A	600V	500 kcmil	THWN/XHHW	600V	200MM ²	EM-IE
420A	600V	600 kcmil	THWN/XHHW	600V	250MM ²	EM-IE
491A				600V	325MM ²	EM-IE

US SYSTEM ENGLISH UNITS EMT, GRS & PVC	US SYSTEM METRIC UNITS EMT, GRS & PVC	CONDUITS SHOWN ON PLANS EMT, GRS & PVC	JAPANESE SYSTEM			JAPANESE SYSTEM HI-PIPE (PVC)
			THIN WALL (EMT)		THICK WALL (GRS)	
			SCREW-LESS	SCREW (THREADED)	SCREW (THREADED)	
1/2 INCH	16 MM	15C	E-19	C-19	G-16	16
3/4 INCH	21 MM	20C	E-25	C-25	G-22	22
1 INCH	27 MM	25C	E-31	C-31	G-28	28
1-1/4 INCH	35 MM	35C	E-39	C-39	G-36	36
1-1/2 INCH	41 MM	40C	E-51	C-51	G-42	42
2 INCH	53 MM	55C	E-63	C-63	G-54	54
2-1/2 INCH	63 MM	65C	E-75	C-75	G-70	70
3 INCH	78 MM	80C	-	-	G-82	82
3-1/2 INCH	91 MM	90C	-	-	G-92	N/A*
4 INCH	103 MM	100C	-	-	G-104	100
5 INCH	129 MM	130C	-	-	-	150



US Army Corps
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 S&P INTERNATIONAL 507 SUMMIT STREET, SUITE 620 HONOLULU, HAWAII 96817	DESIGNED BY:	ISSUE DATE:
	DRAWN BY:	SOLICITATION NO.:
	CHECKED BY:	CONTRACT NO.:
	SUBMITTED BY:	SCALE AS SHOWN
	FILE NAME:	
	ISO A1	

KADENA AIR BASE, OKINAWA, JAPAN
VEHICLE MAINTENANCE SHOP
FY20 MCA PN77833

E-002